The Dynamics Of Latino-Owned Business With Comparisons To Other Ethnicities

by

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CES 16-33 July, 2016

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Abstract

This paper employs the Michigan Census Research Data Center to merge three limited-access Census Bureau data sets by individual firm and establishment level to investigate the factors associated with the Latino-owned Business (LOB) location and dynamics over time. The three main LOB outcomes under analysis are as follows: (1) the probability of a business being Latino-owned as opposed to a business being Asian-owned, Black-owned, or White-owned; (2) the probability of new business entry and exit; and (3) LOB employment growth. This paper then compares these factors associated with LOB with past findings on businesses that are Asian-owned, Black-owned, and White-owned. Some notable findings include: (1) only Black business owners are less associated with using personal savings as start-up capital than Latinos; (2) the only significant coefficient on start-up capital source is personal savings and it increases the odds of survival of a Latino business by 4%; (3) on average, having Puerto Rican ancestry decreases the odds of business survival; and (4) LOB are relatively likely to start a business with a small amount of capital, which, in turn, limits their future growth.

[*] Thanks to Anil Rupasingha, Myriam Quispe-Agnoli, Melissa Banzhaf, J. Clint Carter, Margaret Leventsin, and the Interuniversity Consortium for Political and Social Research for their assistance with the RDC process. Support for this research at the Michigan RDC from NSF (ITR-0427889), Scott Loveridge, the North Central Regional Center for Rural Development, and the Department of Agricultural, Food, and Resource Economics at Michigan State University is also gratefully acknowledged. Any opinions and conclusions expressed herein are those of the author and do not necessarily represent the views of the U.S. Census Bureau. All results have been reviewed to ensure that no confidential information is disclosed.

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Introduction

Entrepreneurship has always been an integral part of the United States economy, but the growth in the rate of immigration to Western countries post World War II and the growth in the rate of small and medium size enterprises (Light and Rosenstein 1995) combine to provide evidence of a revival in entrepreneurship over the past forty years. Small businesses and microenterprises (defined as firms with less than five employees) are very important to the United States economy. They account for approximately 18 percent of employment and create 900,000 jobs a year in the United States (Ramirez de Miess 2006). In the United States, the increased importance of micro and small enterprises is occurring in the context of a rapidly growing Latino population. In many rural areas, the economic impact of Latino immigration is significant and positive (Coates and Gindling 2010). Given the turbulent record of unemployment of the past decade, the possible employment benefits from a more engaged Latino community are substantial.

Researchers explored immigrant and ethnic business ownership extensively over the past 20 years, but small sample size and a lack of micro data on immigrant and ethnic business ownership hindered the success of studies. Robles and Cordero-Guzmán (2007) assess the current literature and write that one of the problems Latino business ownership researchers have is "the lack of a sufficiently large data sample for specific country of origin analysis." This is especially true for relatively sparsely populated rural areas. In fact, studies seeking to look at specific questions about Latino immigrant business ownership in rural areas have been limited in general by small sample size. Studies using microdata (e.g., Bates and Lofstrom 2009), faced limitations in their examination of comprehensive demographic and geographic differences with respect to the success of immigrant business owners in the United States.

Many theories have been proposed regarding the impact of immigrant business owners on their environment and the impact of their environment on their success. For example, the disadvantage theory of entrepreneurship suggests that limited opportunities in the labor market (e.g. from language barriers, discrimination, or unemployment) cause people to seek self-employment (Light 1979, 1980). The structuralist theory proposes that immigrant entrepreneurship experience cannot result entirely from culture, but rather also from the interaction of the group characteristics (e.g. human capital) of different immigrants and the opportunity structures they experience (Waldinger 1993; Waldinger et al. 2000). These theories are obviously very broad and more recently researchers have attempted to examine more specific areas of interest. For example, researchers interested in immigrant business ownership have examined the impact of acculturation, i.e., the process of adapting to a new culture (e.g., Calo 1995), the impact of financial resources (or lack thereof) (e.g., Cavalluzzo and Wolken 2005; Fairlie and Robb 2008b), human and social capital (e.g., Hansen 1995; Peters and Brush 1996), networks (e.g., Walton-Roberts and Hiebert 1997), and motivation, business strategy and community resources (e.g., Liu 2012).

This paper seeks to identify and determine factors associated with the dynamics of Latino-owned business measured in terms of business start-ups, growth, and closure using firm/establishments and employment. Further, this paper compares these factors associated with Latino-owned business with businesses owned by Asians, Blacks, and Whites.

From an economic perspective, it is important to know the factors associated with Latino-owned business (LOB) compared to the factors associated with non-LOB or other minority-owned businesses. Indeed, Martinez and Avila (2011) show that LOB businesses are significantly more likely to go out of business compared to White-owned businesses, so what

causes LOB to exit the market and what are the main barriers to LOB growth? The environment that facilitates entrepreneurship in general and LOB in particular is largely dependent on these factors and on activities of local economic development decision makers and practitioners that promote entrepreneurship. Efforts to promote LOB at various levels of government and nongovernmental entities are dependent upon decision makers and practitioners understanding the determinants of location and growth of LOB, and the characteristics of business owners, businesses, and localities. Understanding these determinants may facilitate developing appropriate guidelines and training of economic development practitioners that will lead to fostering LOB activities more efficiently and effectively.

This paper uses limited-access Census microdata from the Survey of Business Owners, Longitudinal Business Database, and Integrated Longitudinal Business Database to first estimate ethnicities' relative associations with individual, business, and location factors by using a multinomial logit model. Next, the paper examines how these various factors affect the odds of survival of a LOB using Cox survival analysis. Finally, the paper examines how these various factors affect the employment growth of a LOB over time using Pooled OLS. Taken together, these three examinations help paint a more complete picture of how individual, business, and location factors impact LOB survival and growth. Among this paper's conclusion are: (1) Black and Latino business owners are less associated with using personal savings as start-up capital than other ethnicities; (2) use of personal savings increases the odds of survival of LOB by 4%; and (3) relative to other ethnicities, LOB are likely to start a business with a small amount of capital, which, in turn, limits their future growth.

Literature Review

Although studying factors associated with small business has dominated the literature in general, there is a dearth of studies that investigate business and business-owner characteristics associated with LOB in the U.S., primarily because of difficulties obtaining access, using and reporting results from the data. Fairlie and Robb (e.g. 2005a, 2005b, 2006, 2008a) conducted studies on gender differences in business ownership and on minority-owned business using Census RDC data, but those studies have not focused on LOB.

Previous literature found that a major factor impacting Latino business survival and growth is the lack of financial resources, and that Latinos primarily finance their businesses with personal savings and informal loans from friends or family and moneylenders (Granier 2006, Haynes, Onochie, and Lee 2008, Raijman and Tienda 2000). Similarly, data from the 2005 National Minority Business Owner Survey suggests that in comparison to Korean-Americans, Mexican-American business owners borrow more from family, friends, suppliers, and credit cards and have a lower proportion of bank loan debt (Haynes, Onochie, and Lee 2008). The findings of Martinez et al. (2011) contradicted those of previous studies (Haynes, Onochie, and Lee 2008, Cavalluzzo and Wolken 2005, Granier 2006, and Blanchard, Zhao, and Yinger 2008) on the types of startup funds that Latino business owners use, as they did not find that Latino business owners use fewer formal funds compared to Whites. Indeed, Martinez et al. (2011) find that African Americans were the only race group significantly less likely to use formal funds compared to Whites. These results support the findings of Blanchard, Zhao, and Yinger (2008), which show that Latinos receive higher interest rates on bank loans, which also discourages them from seeking formal funds for their businesses.

There have not been many studies into the interaction of gender and minority business factors, though Martinez and Avila (2011) use data from the Kauffman Foundation to show that neither Latinos nor Latinas differ significantly from White men or women in terms of having a college degree. Martinez and Avila (2011) also find that Latinos owned significantly fewer home-based businesses compared to White men.

Recent studies attempted to examine why Latinos tend to concentrate in sectors perceived as relatively vulnerable, such as the services (Puryear et al. 2008, Robles and Cordero-Guzmán 2007), construction, wholesale trade, and retail trade sectors (U.S. Census Bureau 2010). Martinez et al. (2011) explain LOB are often associated with low-barrier industries because they lack financial capital to enter high-barrier industries. High barrier industries are those requiring advanced educational credentials or large amounts of startup capital. Industries such as professional services, finance or insurance are examples of high barrier industries while low barrier industries include some food services and construction.

Robb & Coleman (2009) conclude that women business owners also have more difficulty keeping their businesses in operation. Bates, Lofstrom, and Servnon (2010) find that businesses within the low technology sectors or low barrier industries face a greater likelihood of going out of business compared to businesses within the high technology sectors or the high barrier industries. Martinez and Avila (2011) also find that Latina-owned businesses represent a larger percentage of businesses within the low technology sector and a smaller percentage of businesses within the medium and high technology sectors compared to White-owned businesses, but not that Latino-owned businesses were concentrated in the low technology sectors compared to White men-owned businesses, or represented a significantly smaller percentage of businesses within the high or medium technology sectors. These results contradict Bates, Lofstrom, and

Servnon (2010). Other factors found to be associated with higher LOB survival rates are older owner age (Fairlie 2005) and higher education (Martinez and Avila 2011).

Methods

The literature shows numerous testable hypotheses. Thus this paper will test each of the following seven hypotheses:

H1. Latino business owners primarily finance their businesses with personal savings.

H2. Latino business owners are less likely to finance their business with bank loans.

H3. Latino business owners do not differ significantly from White business owners in their likelihood of having a college degree.

H4. LOB are more associated with low-barrier industries such as the services, construction, wholesale trade, and retail trade sectors.

H5. LOB are less associated with high-barrier industries such as professional services, and finance or insurance.

H6. Latina-owned businesses are less likely to survive than Latino-owned businesses.

H7. LOB in low-barrier industries are less likely to survive.

The empirical approach to implement the aforementioned objectives and test these hypotheses is grounded in the theory of firm entry, growth and exit, which is well-developed in the small business and industrial organization literature (Borjas 1986; Evans 1987; Evans and Jovanovic 1989; Evans and Leighton 1989; Dunne, Roberts and Samuelson 1989; Dunne and Hughes 1994). I draw on this literature to motivate the behavior of LOB. A generalized version

of this framework allows the model to explain LOB activity or outcomes as a function of explanatory factors as well as stochastic variables as follows:

$$(1) \qquad y_{it} = f(X_{1t}, \ldots, X_{it}, \varepsilon_i)$$

where y_{it} is the LOB outcome variable at time t for business i, X_i's are the explanatory factors that affect the outcome, and ε_i is a random error term. This approach can capture business location and dynamics (firm entry, growth and exit behavior). This model allows incorporation of business, business owner, and region-specific factors such as business size and age, industry, access to capital, demographic, economic, and location specific factors such as population density (including density of populations from various country of ancestral origins), amenities, local government finance, and distance-to-urban centers.

The three main LOB outcomes in the analysis are as follows:

1. The probability of a business being Latino-owned as opposed to a business being Asian-owned, Black-owned, or White-owned.
2. The first LOB dynamic outcome: the probability of LOB year-over-year survival.
3. The second LOB dynamic outcome: LOB employment growth factors.

In the analysis of the first LOB outcome, I use a multinomial logit model to estimate the factors associated with LOB, Asian businesses, and businesses owned by Blacks and by Whites. The racial/ethnic groups of businesses in the data is categorical (taking on non-ordered outcome values for each type of ownership such as LOB, Asian-owned, Black-owned, White-owned).[3] The model assesses how these different owner groups are related to personal traits and

[3] For the sake of this study, ethnicities are assumed exclusive (e.g., if someone if Hispanic and White, they appear as Hispanic in the multinomial). While this assumption overlooks detail in the nonexclusive nature of ethnicity, the model nonetheless provides an examination of factors associated with each ethnic group.

demographic variables, business and industry characteristics, and location factors. I estimate multinomial logit model as follows (Greene 2000):

$$(2)\ \Pr(y = k) = \frac{\exp(\propto + \beta' X_{ij})}{\sum_{j=1}^{3} \exp(\propto + \beta' X_{ij})}\ k = 1,2,3$$

A categorical dependent variable be defined so that it takes on four levels (0 for LOB, 1 for Asian-owned businesses, 2 for Blacks-owned businesses, and 3 for White-owned businesses). The vector X denotes the set of characteristics associated with the businesses. Thus this examination will use LOB as the base group and the results will be presented in terms relative to LOB. Often the omitted group in similar analyses is White individuals, and the relative position of Latino individuals must be inferred. As Latino business owners are the focus of this paper, they will be the base group in this analysis and hence all ethnic associations will be in terms of their association relative to Latino business owners.

Note that the explanatory variables that represent personal traits refer to the traits of the owner or majority shareholder of the firm and include variables such as whether the owner had prior experience owning a business, and hours per week spent working.[4] Demographic information includes traits such as gender, education, disability, marital status, and age, and country of ancestral origin. Personal and demographic traits of business owners, business age, and sources of capital are available from the SBO. Business characteristics such as firm age, size, and industry can be constructed using information available from the merged ILBD/LBD.

Regional factors include demographic information of the county including population shares from Latin American countries, local unemployment rates or some measure of local conditions, and the rurality of U.S. counties. The rurality of U.S. counties follows the

[4] If individuals share ownership of a firm equally, I assign ownership to one of the owners at random.

contemporaneous USDA Urban-Rural Continuum Codes. Demographics of locality include measures such as the racial mix, the age distribution, the proportion of the local population with the same country of ancestral origin, and the proportions of the local population speaking English, etc. These county level characteristics are publicly available from the decennial census and Bureau of Economic Analysis and are merged with the ILBD/LBD and SBO data using business location information.

Regional factors, including demographic information of the county, may include interactions between the country of ancestral origin of the firm owner and the demographics of the locality in some specifications, allowing us to answer my research questions regarding the impact of Latino population density (especially Latino with the same ancestral origin) on business development outcomes. Similarly the coefficients on the owner's country of ancestral origin variables help investigate the impact of country of origin on firm survival.

This paper uses duration analysis employing the hazard model to investigate the determinants of new firm entry and exit (or more specifically firm survival). This framework allows us to quantify the timing of the exit, rather than the mere incidence. The application of a hazard model (Cox 1972, 1975) for likelihood of firm survival has become a routine method for survival data used in the empirical literature since Audretsch and Mahmood (1995; see Musso and Schiavo 2008 and Christie and Sjoquist 2012 for recent applications). Following the standard notation of survival analysis models, let *T* denote the duration of time that a firm has been in existence and *t* as the current time. The probability that this firm does not survive within a short interval of time, $(t, t + dt)$, conditional on the fact that it has survived as of period *t* can be calculated as follows:

$$(3)\ h_i(t|X) = \lim_{dt \to 0} \frac{P(t \leq T \leq t + dt | T \geq t; X_i)}{dt}$$

where X is the vector of covariates that are likely to affect the duration of the firm. Empirically, to examine the effects of the covariates on the duration, researchers have used a proportional hazards model (Cox 1972) which assumes the hazard rate is a multiplicative function of a baseline hazard, $h_0(t)$, and an exponential function of a set of covariates (Christie and Sjoquist, 2012):

$$(4)\ h_i(t|X_i) = h_0(t)\ exp(X_i\beta)$$

where β represents the usual vector of coefficients, and X_i is a vector of explanatory variables. In addition to above explanatory factors, following firm survival literature, I also include in this model the firm age, employment, and payroll. This paper uses the Cox proportional hazard model to estimate the hazard rates and the conditions affecting them. The primary focus of this exercise is on variables associated with LOB survival. The hazard function gives the risk that an establishment does not survive in the next year, on the condition that this business has survived up to the beginning of the current year (Christie and Sjoquist 2012). Length of survival is calculated as the number of years a business reports positive employment from the start-up year.

The probability of survival is assumed to be dependent on the same set of explanatory factors used in equation 2. Since there is little theoretical support for any particular parametric shape of the baseline hazard, most studies use a nonparametric approach (Christie and Sjoquist 2012) for estimation. Therefore, I use a nonparametric approach, but also use parametric models in order to check whether my results are sensitive to the choice of a particular assumption regarding the distribution.

I use definitions and methods provided by the Bureau of the Census Business Dynamics Statistics website[5] to formulate employment change variables. The firm- or establishment–level employment growth rate is defined as follows:

$$(5)\quad g_{it} = \frac{(E_{it} - E_{i,t-\tau})}{X_{it}} \quad where\ X_{it} = \frac{1}{2}(E_{it} + E_{i,t-\tau})$$

E_{it} is employment in year *t* for establishment *i* and $E_{i,t-\tau}$ is employment in year $t - \tau$ for establishment *i*. Known as DHS measure (Haltiwanger et al. 2009), this measure can be defined at any level of aggregation (establishment, firm, local area, industry, etc.) and details of calculations including measures of job creation and destruction at the establishment and firm level are given in Haltiwanger et al. (2010). I use the establishment-level growth rate measures and the estimation equation takes the following form:

$$(5)\quad g_{it} = f(\beta X_{it}, \gamma Z_{it}, \delta L_{it})$$

where g_{it} refers to employment growth for the ith firm, X is a vector of personal traits and demographic variables, Z is a vector of firm and industry characteristics, and L is a vector of regional variables. I estimate this model using linear relationships. First, I use a 1-year time horizon (i.e., $\tau = 1$) to calculate the employment growth. Then I repeat the model with 2 and 3-year time intervals (i.e., $\tau = 2, 3$) to test the stability of the initial specification over time.

Data

This paper merges three Census Bureau data sets by individual firm and establishment level to investigate the factors associated with LOB location and dynamics over time. The three

[5] See http://www.census.gov/ces/dataproducts/bds/overview.html for more details.

databases are the Integrated Longitudinal Business Database (ILBD, $n =$ approximately 18 million annually), the Longitudinal Business Database (LBD, $n =$ approximately 8 million annually) and the Survey of Business Owners (SBO, $n =$ approximately 2.3 million in 2007). The SBO is not a longitudinal survey of businesses. However, by linking the SBO with the LBD and the ILBD by firm id number, one can trace longitudinal changes of the businesses that are matched.

Access to these three micro-datasets is limited. Qualified researchers with approved Federal Statistical Research Data Center (RDC) projects gain restricted access to the data at one of the 22 secure RDC locations and are sworn for life to protect the confidentiality of the data they access. The Center for Economic Studies (a unit within the Census) considers proposals from qualified researchers consistent with the subject matter of the surveys and censuses collected by the Census Bureau. Most economic datasets also contain Federal Tax Information and thus also require approval from the Internal Revenue Service. The application process is lengthy and includes a background check as a prerequisite to gaining Special Sworn Status. Researchers must receive Special Sworn Status before gaining access to Census microdata. Moreover, these researchers are subject to legal obligations and penalties should they violate the confidentiality agreement. Approved projects gain access to only those datasets approved within their project and all results must go through disclosure analysis to verify that they contain no risk of improper disclosure of any identifiable information. In the analysis that follows, certain estimated coefficients are non-disclosed to protect confidentiality of individuals and businesses. Under Census guidelines, the reader can be shown that a non-disclosed variable was included in estimated equation, but not the coefficient value, sign, or significance.

Factors associated with LOB ownership and dynamics include characteristics of business and industry, characteristics of business owner, and characteristics of location. The standard economic model predicts that these factors are associated with the firm's production process. While measures of business ownership are available from the SBO, the data are cross-sectional and cannot capture the dynamics of LOB. Measures of business creation, expansion, and destruction are needed to represent dynamics of businesses. Such measures are available from the LBD or can be created using the LBD. On the other hand, the LBD lacks information on the characteristics of business owners such as race, ethnicity, gender, education level, whether the owner is foreign born, and access to capital. Such information is available from the SBO.

The LBD has firm and establishment identifiers, making the linking of the LBD with the SBO feasible. The ILBD is needed to address the issue that while SBO has information about firms that have no paid employees, the LBD has information only about establishments that have paid employees. The ILBD consists of administrative records for all nonemployer business units. By linking the ILBD, LBD, and SBO, one can track the dynamics of not only employer LOB but also the nonemployer LOB including the transitions of LOB nonemployers to LOB employers. Although location of the businesses are available from these merged data sets, location specific characteristics such as agglomeration, racial make-up of location, amenities, and market size are not available in this data. This requires linking merged data with location specific factors that are publicly available from the Bureau of the Census, the BEA and the USDA.

The first objective of this paper is to study the factors associated with LOB in comparison to the businesses owned by Asians, Blacks and Whites. The model identifies these businesses using 2002 SBO. Because we are interested in age and initial size (using start-up employment), I

merge the 2002 and 2007 SBO data with pre-2008 ILBD/LBD data sets that date back to 1994 for some businesses.

The second objective of this paper is to study the factors associated with LOB dynamics, measured using business duration and employment growth. First, I measure the survival of LOB establishments. This requires creating a cohort of only LOB. Because a LOB's inclusion in the 2007 SBO implies its prior survival, for this objective I only use the 2002 SBO and examine the time period from 2002-2007. The merged 2002 SBO with ILBD/LBD facilitate tracking these businesses for the interim period (2002-2007) with business characteristics. Given the panel nature of the created data set, it is possible to track the survival rates of cohorts over the time period 2002-2007. Secondly, I measure employment growth of these LOB over time. I use a one-year time horizon to calculate the employment growth first and then experiment with expanded time horizons such as 2-year and 3-year time intervals to test the stability of the initial specification.

Results

The results of the multinomial logit analysis of the first LOB outcome – the probability of a business being Latino-owned as opposed to a business being Asian-owned, Black-owned, or White-owned – is given in Table 5.[6] For ease of interpretation, Table 5 displays the results as relative risk ratios (or odds ratios, i.e., as exponentiated multinomial logit coefficients). Table A1 in the appendix contains the coefficients on control variables in the estimation that are not included in Table 5. These variables are listed below:

[6] Note that "XX" may appear rather than a coefficient and standard error in the tables using limited access data if that coefficient and standard error have been suppressed upon release of the results because the Census Bureau finds that coefficient and standard error to risk an inappropriate disclosure of individual response.

Establishment-Level Variables

- Business organization type indicators:
  - Coop
  - Estate
  - Nonprofit
  - Public
  - Husband and wife
- Establishment employment
- Establishment payroll
- Firm employment
- Firm payroll
- State industry employment
- State indicators variables

County-Level Variables

- Population
- Population squared
- Population density
- Age 25 to 34 population
- Age 35 to age 44 population
- Age 45 to 54 population
- Age 55 to 64 population
- Age greater than 64 population
- Female population
- Female population squared
- Black population
- Black population squared
- Asian population
- Asian population squared
- Hispanic population
- Hispanic population squared
- Number of build permits issued
- Industry employment

Though some of the coefficients on the county-level control variables are not statistically significant, most are statistically significant, but equal to 1.00. Equality to 1.00 implies that (*ceteris paribus*) the represented ethnic group is near equal to Hispanic-Americans[7] in its relationship with the control variable in question. The employment and payroll variables relate similarly. The logit coefficients are given in the appendix Table A2.[8]

Table 5. Multinomial Logit Results (Relative Risk Ratio)

	(1)	(2)	(3)

[7] The paper uses the term Hispanic, rather than Latino, in some cases because the Census data categorizes individuals as Hispanic, rather than Latino. This paper acknowledges the difference between the two terms, but given the population of interest is Latino and that most individuals in the U.S. who are Hispanic are also Latino, this paper assumes results on Hispanic entrepreneurs in the U.S. are be similar to results on Latinos in the U.S.

[8] Numerous coefficients in this and other tables are statistically significant, but displayed as 0.00*** or -0.00***. The Bureau of the Census limits the specificity with which the regression results can be displayed to protect against improper disclosure, so only the sign of the coefficient is available. As these coefficients are near zero, their impact, while statistically significant, is small.

Business Variables	Asian-American	Black-American	White-American
Female	0.92***	1.24***	0.76***
	(0.02)	(0.02)	(0.01)
Ed level: high school	1.66***	2.26***	3.31***
	(0.05)	(0.08)	(0.07)
Some college	XX	3.69***	4.77***
	XX	(0.14)	(0.11)
Trade school	1.29***	3.50***	3.76***
	(0.06)	(0.16)	(0.11)
Associates degree	2.15***	XX	4.64***
	(0.09)	XX	(0.14)
Bachelor's degree	4.67***	3.78***	8.06***
	(0.15)	(0.15)	(0.19)
Graduate degree	5.91***	3.34***	7.55***
	(0.21)	(0.14)	(0.20)
Owner age: 25 to 34	XX	XX	0.11***
	XX	XX	(0.00)
35 to 44	0.26***	0.37***	0.13***
	(0.01)	(0.02)	(0.00)
45 to 54	0.29***	0.49***	0.19***
	(0.01)	(0.03)	(0.01)
55 to 64	0.30***	0.62***	0.27***
	(0.01)	(0.03)	(0.01)
$\geq$ 65	0.24***	0.72***	0.37***
	(0.01)	(0.04)	(0.01)
Hours: 20 to 39 hours	0.78***	XX	0.78***
	(0.02)	XX	(0.02)
40	0.98	0.81***	0.67***
	(0.03)	(0.03)	(0.02)
41 to 59	0.88***	0.89***	0.78***
	(0.03)	(0.03)	(0.02)
$\geq$ 60	1.04	1.27***	0.72***
	(0.03)	(0.04)	(0.02)
Primary income	1.17***	0.59***	0.96**
	(0.03)	(0.01)	(0.02)
Provide good/service	0.79***	XX	0.94***
	(0.01)	XX	(0.01)
Manage	1.10***	1.17***	1.12***
	(0.02)	(0.03)	(0.02)
Financial	1.05***	0.92***	1.48***
	(0.02)	(0.02)	(0.02)

Table 5. (cont'd)

Business Variables	(1) Asian-American	(2) Black-American	(3) White-American
Start-up capital: $5,000-	0.79***	0.94*	0.76***
$10,000	(0.03)	(0.03)	(0.02)
$10,000-$25,000	1.15***	0.82***	0.82***
	(0.03)	(0.03)	(0.02)
$25,000-$50,000	1.32***	0.79***	0.87***
	(0.05)	(0.04)	(0.02)
$50,000-$100,000	XX	0.73***	0.96
	XX	(0.04)	(0.03)
$100,000-$250,000	1.83***	0.75***	1.07**
	(0.07)	(0.04)	(0.04)
$250,000-$1 million	1.85***	0.74***	1.23***
	(0.10)	(0.05)	(0.06)
> $1 million	2.13***	XX	1.45***
	(0.18)	XX	(0.11)
Start-up capital source:	1.08***	0.93***	1.00
Savings	(0.02)	(0.02)	(0.02)
Credit	0.91***	1.11***	0.85***
	(0.03)	(0.03)	(0.02)
Government loan	XX	1.32***	0.89**
	XX	(0.09)	(0.05)
Bank loan	1.29***	0.89***	1.19***
	(0.04)	(0.03)	(0.03)
Venture capital	1.33***	1.06	1.32***
	(0.10)	(0.10)	(0.08)
None Needed	0.82***	0.88***	0.92***
	(0.02)	(0.03)	(0.02)
Expansion capital source:	1.11***	1.06**	0.98
Savings	(0.02)	(0.03)	(0.02)
Credit	0.74***	0.94**	1.09***
	(0.02)	(0.03)	(0.02)
Government loan	0.81***	1.14	0.78***
	(0.06)	(0.10)	(0.05)
Bank loan	0.89***	0.73***	1.12***
	(0.03)	(0.03)	(0.02)
Venture capital	0.94	XX	0.92
	(0.10)	XX	(0.08)
None needed	0.98	0.83***	1.20***
	(0.02)	(0.02)	(0.02)

Table 5. (cont'd)

Business and County Variables	(1) Asian-American	(2) Black-American	(3) White-American
Multiple establishments	0.68***	1.12**	2.00***
	(0.03)	(0.05)	(0.07)
>10% sales: Federal	1.09*	1.07	0.66***
	(0.05)	(0.06)	(0.03)
State or local gov't	0.67***	1.40***	0.89***
	(0.03)	(0.05)	(0.02)
Primary customer: other	0.90***	1.11***	0.89***
businesses	(0.01)	(0.01)	(0.01)
Individuals	1.02*	0.90***	1.05***
	(0.01)	(0.01)	(0.01)
County-level variables:			
Per capita income	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
Mean household size	0.60***	0.66***	0.22***
	(0.04)	(0.05)	(0.01)
% of hh headed by single female	4.56***	27,791.40***	0.01***
	(1.94)	(13,190.53)	(0.00)
Unemployment rate	17.58***	2.03	1.62
	(14.48)	(1.91)	(1.03)
Labor force participation rate	1.53*	0.68	3.39***
	(0.35)	(0.17)	(0.57)

Table 5. (cont'd)

NAICS Sector Industry	(1) Asian-American	(2) Black-American	(3) White-American
11: Ag., Forestry, Fishing, Hunting	0.66*	0.59**	1.87***
	(0.14)	(0.12)	(0.30)
21: Mining, Quarrying, Oil & Gas	0.50***	0.56**	3.79***
Extraction	(0.13)	(0.16)	(0.71)
22: Utilities	0.98	1.27	2.08***
	(0.34)	(0.41)	(0.45)
23: Construction	0.37***	0.58***	1.43**
	(0.07)	(0.10)	(0.20)
31-33: Manufacturing	1.08	0.42***	1.57***
	(0.19)	(0.08)	(0.22)
42: Wholesale Trade	1.96***	0.40***	1.64***
	(0.35)	(0.07)	(0.23)
44-45: Retail Trade	1.98***	0.62***	1.37**
	(0.34)	(0.11)	(0.19)
48-49: Transportation and	0.74*	XX	0.86
Warehousing	(0.13)	XX	(0.12)
51: Information	0.92	1.17	1.95***
	(0.16)	(0.22)	(0.28)
52: Finance and Insurance	0.80	0.89	1.76***
	(0.14)	(0.16)	(0.25)
53: Real Estate, Rental, Leasing	1.25	0.73*	XX
	(0.22)	(0.13)	XX
54: Professional, Scientific, and	0.98	0.85	1.66***
Technical Services	(0.17)	(0.15)	(0.23)
55: Management of Companies	1.65**	1.14	1.67**
and Enterprises	(0.40)	(0.30)	(0.33)
56: Administrative Support, Waste	0.47***	0.83	0.88
Management & Remediation Services	(0.08)	(0.15)	(0.12)
61: Educational Services	0.97	1.00	1.37**
	(0.18)	(0.19)	(0.20)
62: Health Care and Social	1.09	1.41*	0.96
Assistance	(0.19)	(0.26)	(0.14)
71: Arts, Entertainment, & Recreation	0.62***	1.46**	2.08***
	(0.11)	(0.26)	(0.30)
72: Accommodation & Food Services	3.07***	0.44***	0.71**
	(0.54)	(0.08)	(0.10)
81: Other Services	XX	1.01	1.14
	XX	(0.18)	(0.16)

Table 5. (cont'd)

County Variables	(1) Asian-American	(2) Black-American	(3) White-American
Amenities	0.84***	0.75***	0.86***
	(0.01)	(0.01)	(0.01)
Rural-urban code 2	1.01	1.08**	1.13***
Metro area, 0.25 to 1 million pop.	(0.03)	(0.04)	(0.03)
Rural-urban code 3	0.91**	1.05	1.28***
Metro area, < 250,000	(0.04)	(0.05)	(0.04)
Rural-urban code 4	0.89**	1.11	1.36***
Adjacent to metro, ≥ 20,000	(0.05)	(0.07)	(0.06)
Rural-urban code 5	0.89	1.04	1.58***
Not adjacent to metro, ≥ 20,000	(0.07)	(0.10)	(0.10)
Rural-urban code 6	0.89*	1.22***	1.76***
Adjacent to metro, 2,500-19,999	(0.06)	(0.08)	(0.09)
Rural-urban code 7	0.75***	0.83**	1.54***
Not adjacent to metro, 2,500-19,999	(0.06)	(0.07)	(0.09)
Rural-urban code 8	0.74*	1.59***	1.94***
Adjacent to a metro, < 2,500	(0.13)	(0.25)	(0.25)
Rural-urban code 9	0.37***	1.11	1.92***
Not adjacent to a metro, < 2,500	(0.06)	(0.17)	(0.21)
Constant	0.10***	0.01***	100.23***
	(0.05)	(0.00)	(36.10)
State FE	YES	YES	YES
Year FE	YES	YES	YES

Pseudo R-squared = 0.224
n = ~11,847,500[9]
Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

The results presented in Table 5 show, ceteris paribus, the relative risk ratio of business owner ethnicity relative to Hispanic-American associated with various factors. For example, Table 5 shows that if a business owner is female, relative to Hispanic-American, the odds that she is Asian-American change by a factor of 0.92, 1.24 for Black-American, and 0.76 for White-American. The coefficients on owner education indicate that Hispanic-American business owners are less likely to be associated with any level of education included. Thus Hispanic-

[9] The exact number of observations is suppressed by the Bureau of the Census to avoid improper disclosure of an individual response.

American business owners are more likely to be associated with the omitted education level: less than a high school degree. Conversely, for the age group coefficients, Hispanics-American business owners are more likely to be associated with any of the included age groups. Thus Hispanic-American business owners are less associated with the less than 25 years old category.

Hispanic-American business owners' relative association with weekly hours worked is more complex. Hispanic-American business owners are more likely to work 20 to 59 hours per week on their business; they are also more likely than White-Americans to work 60 or more hours per week, but less likely than Asian and Black-American business owners to work 60 or more hours per week. Hispanic-American business owners are more likely to be associated with having the business as their primary source of income than both Black and White-Americans, but less likely than Asian-Americans.

The start-up capital amount is divided into value ranges (imposed by the 2007 SBO). The omitted category is less than $5,000. Hispanic-Americans are more likely to be associated with the $5,000 to $10,000 range than the other ethnicities. Hispanic-Americans are more likely to be associated with the ranges in $10,000 to $100,000 than White and Black-Americans, but less likely than Asian-Americans (though the $50,000 to $100,000 range is suppressed for Asian-Americans). Hispanic-Americans are less associated with all of the ranges over $100,000 than Asian and White-Americans, but more associated with those (non-suppressed) ranges than Black-Americans. Start-up and expansion capital source associations have statistically significant variation.

The county-level variables for percent of the population with various education levels show that Hispanic-American business owners are less associated with every included education level. Hence, Hispanic-American business owners tend to be located in counties with a relatively

high percent of individuals having less than a high school degree. The coefficients on other county-level variables reveal that LOB are more associated with areas in which there are high amenities than any other ethnicity. The amenities variable uses the USDA ERS amenities scale, which is constructed by combining six measures of climate, topography, and water area that reflect environmental qualities most people prefer. Further, the coefficients on the rural-urban continuum codes indicate that LOB are more associated with rural counties than Asian-American business owners, but less associated with rural counties than White-Americans.

Table 5 shows that Asian-American business owners are more likely than Hispanic-American business owners to be located in counties with higher unemployment rates. White and Black-American business owners do not have a statistically significant difference from Hispanic-Americans in their association with the unemployment rate. The results also show many statistically significant results surrounding the various NAICS 2-digit industries. Most of those results are left to the reader to examine, but note that if a businesses is in the construction and agricultural industries, two industries often associated with Hispanic-Americans, the odds of it being owned by a White-American rather than a Hispanic-American increase by 1.43 and 1.87 times, respectively.

The results of the analysis into the factors associated with a LOB's survival probability are given in Table 6. The hazard function gives the risk of survival, that is, the probability that a business will still be in business in the next year, on the condition that this business has survived up to the beginning of the current year (Christie and Sjoquist 2012). As noted above, there is little theoretical support for any particular parametric shape of the baseline hazard and, as a result, most studies use a non- or semi-parametric approach (Christie and Sjoquist 2012) for estimation. Therefore, the paper focuses on the results given by the Cox proportional hazard

model, but also includes two parametric models – exponential and Weibull – to check the sensitivity of the results to the choice of a particular assumption regarding the distribution. Table 6 contains the results of these three models and shows that the results are stable across model selection. For ease of comparison, these results are presented as hazard ratios.

For readability, Table A3 in the appendix contains the coefficients on variables that are controlled, but not included in Table 6.[10] Equality to 1.00 implies that (*ceteris paribus*) as the variable in question increases by 1, the odds that the business will survive in the next year decrease by 1.00 times (i.e. remain about the same). The Cox Proportional Hazard Model, exponential, and Weibull coefficients (the parametric results are presented in accelerated failure time form) are given in the appendix Table A4.

[10] These variables include: business organization type indicators; establishment employment; establishment payroll; firm employment; firm payroll; state-level industry employment; and state indicator variables. Table A3 also contains some of the county-level control variables: population, population squared, population density, age group populations; female, Black, Asian, Hispanic, Mexican, Cuban, and Puerto Rican population and populations-squared; number of build permits issued; industry employment; Mexican, Cuban, and Puerto Rican employment and establishments; and interaction terms between ancestral origin and local population and number of business owners of the same origin. Though some of the coefficients on these county-level control variables are statistically significant, almost all are significant and equal to 1.00.

Table 6. Cox Proportional Hazard, Exponential, Weibull Survival Model Results (Odds Ratio)

Business Owner Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Female	1.06***	1.06***	1.06***
	(0.02)	(0.02)	(0.02)
Mexican	1.02	1.02	1.02
	(0.03)	(0.03)	(0.03)
Cuban	1.00	0.99	0.98
	(0.04)	(0.04)	(0.04)
Puerto Rican	1.19***	1.20***	1.22***
	(0.05)	(0.05)	(0.06)
Owner education: High school	0.99	1.00	1.01
	(0.04)	(0.04)	(0.04)
Some college	1.00	1.00	1.02
	(0.04)	(0.04)	(0.04)
Tech or trade school	0.96	0.96	0.97
	(0.05)	(0.05)	(0.05)
Associates degree	0.98	0.99	1.00
	(0.06)	(0.06)	(0.06)
Bachelor's degree	1.01	1.01	1.03
	(0.04)	(0.04)	(0.04)
Graduate degree	0.96	0.97	1.00
	(0.04)	(0.04)	(0.05)
Business Age	1.02	0.99***	0.91***
	(0.02)	(0.00)	(0.00)
Owner age: 25 to 34	0.89*	0.91	0.95
	(0.06)	(0.06)	(0.06)
35 to 44	0.81***	0.81***	0.79***
	(0.05)	(0.05)	(0.05)
45 to 54	0.82***	0.81***	0.78***
	(0.05)	(0.05)	(0.05)
55 to 64	0.89**	0.89**	0.87**
	(0.05)	(0.05)	(0.05)
Older than 64	0.95	0.96	1.00
	(0.06)	(0.06)	(0.06)
Hours: 20 to 39 hours	0.94	0.95	0.97
	(0.04)	(0.04)	(0.05)
40	0.95	0.96	0.98
	(0.04)	(0.04)	(0.04)
41 to 59	0.86***	0.86***	0.87***
	(0.04)	(0.04)	(0.04)
More than 59	0.89**	0.90**	0.92*
	(0.04)	(0.04)	(0.04)

Table 6. (cont'd)

Business Owner Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Primary income	0.85***	0.84***	0.82***
	(0.02)	(0.02)	(0.02)
Provide goods/services	1.04	1.04	1.05*
	(0.03)	(0.03)	(0.03)
Management	1.03	1.02	1.02
	(0.03)	(0.03)	(0.03)
Finances	0.94**	0.94**	0.94**
	(0.02)	(0.02)	(0.02)
Multiple establishments	0.75***	0.75***	0.75***
	(0.03)	(0.03)	(0.03)
State-up capital source: Savings	0.94**	0.94**	0.93**
	(0.03)	(0.03)	(0.03)
Assets	1.00	1.00	1.00
	(0.03)	(0.03)	(0.03)
Credit	0.94	0.95	0.96
	(0.04)	(0.04)	(0.04)
Government loan	0.98	0.98	0.99
	(0.07)	(0.07)	(0.07)
Bank loan	0.99	0.99	1.00
	(0.03)	(0.03)	(0.03)
Venture capital	1.04	1.04	1.05
	(0.06)	(0.06)	(0.06)
None needed	1.06	1.07	1.09*
	(0.05)	(0.05)	(0.05)
Expansion capital source: Savings	1.10***	1.11***	1.13***
	(0.04)	(0.04)	(0.04)
Assets	1.06	1.07	1.09*
	(0.05)	(0.05)	(0.05)
Credit	1.06	1.05	1.05
	(0.04)	(0.04)	(0.04)
Government loan	1.00	0.99	0.99
	(0.08)	(0.08)	(0.08)
Bank loan	0.81***	0.80***	0.78***
	(0.03)	(0.03)	(0.03)
Venture capital	1.09	1.10	1.13
	(0.08)	(0.08)	(0.09)
None needed	0.96	0.96	0.96
	(0.03)	(0.03)	(0.04)

Table 6. (cont'd)

Business Owner and County Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
>10% sales: federal government	1.10*	1.10*	1.10*
	(0.06)	(0.06)	(0.06)
State or local government	0.84***	0.84***	0.84***
	(0.04)	(0.04)	(0.04)
Primary customer: other businesses	0.93***	0.93***	0.93***
	(0.02)	(0.02)	(0.02)
Individuals	0.85***	0.85***	0.85***
	(0.02)	(0.02)	(0.02)
County-level variables:			
% with education: High school	2.56	2.46	2.14
	(1.58)	(1.52)	(1.35)
Some college	5.62**	5.65**	6.19**
	(3.89)	(3.92)	(4.39)
Associates degree	1.94	1.80	1.35
	(2.28)	(2.12)	(1.63)
Bachelor's degree	1.49	1.44	1.24
	(1.00)	(0.97)	(0.85)
Graduate degree	2.40	2.31	2.13
	(1.87)	(1.80)	(1.69)
Per capita income	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Unemployment rate	2.28	2.09	1.65
	(2.36)	(2.16)	(1.75)
Labor force participation rate	1.00	0.99	0.96
	(0.33)	(0.33)	(0.33)

Table 6. (cont'd)

NAICS Sector Industry	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
21: Mining, Quarrying, Oil & Gas Extraction	0.97	1.00	1.11
	(0.21)	(0.22)	(0.25)
22: Utilities	XX	XX	XX
	XX	XX	XX
23: Construction	0.92	0.95	1.04
	(0.10)	(0.10)	(0.12)
31-33: Manufacturing	0.92	0.94	0.99
	(0.10)	(0.10)	(0.11)
42: Wholesale Trade	0.82*	0.84	0.90
	(0.09)	(0.09)	(0.10)
44-45: Retail Trade	0.94	0.97	1.04
	(0.10)	(0.11)	(0.12)
48-49: Transportation and Warehousing	0.95	0.97	1.04
	(0.10)	(0.11)	(0.12)
51: Information	XX	XX	XX
	XX	XX	XX
52: Finance and Insurance	XX	XX	XX
	XX	XX	XX
53: Real Estate and Rental and Leasing	0.80*	0.82*	0.89
	(0.09)	(0.10)	(0.11)
54: Professional, Scientific, and Technical Services	0.85	0.86	0.90
	(0.09)	(0.09)	(0.10)
55: Management of Companies and Enterprises	1.09	1.11	1.16
	(0.19)	(0.20)	(0.21)
56: Administrative Support and Waste Management and Remediation Services	0.93	0.95	1.01
	(0.11)	(0.11)	(0.12)
61: Educational Services	0.72**	0.73**	0.75**
	(0.10)	(0.10)	(0.11)
62: Health Care and Social Assistance	0.69***	0.71***	0.75**
	(0.08)	(0.09)	(0.09)
71: Arts, Entertainment, and Recreation	0.92	0.94	1.00
	(0.12)	(0.12)	(0.13)
72: Accommodation and Food Services	0.82*	0.84	0.89
	(0.09)	(0.09)	(0.10)
81: Other Services	0.85	0.87	0.92
	(0.09)	(0.10)	(0.10)
92: Public Administration	XX	XX	XX
	XX	XX	XX

Table 6. (cont'd)

County Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Amenities	XX	XX	XX
	XX	XX	XX
Rural-urban code 2	1.06	1.06	1.06
Metro area, 0.25 to 1 million pop.	(0.04)	(0.04)	(0.04)
Rural-urban code 3	0.97	0.98	0.98
Metro area, < 250,000	(0.05)	(0.05)	(0.06)
Rural-urban code 4	1.01	1.01	1.01
Adjacent to metro, ≥ 20,000	(0.07)	(0.07)	(0.08)
Rural-urban code 5	1.00	1.00	0.99
Not adjacent to metro, ≥ 20,000	(0.10)	(0.10)	(0.10)
Rural-urban code 6	0.95	0.94	0.93
Adjacent to metro, 2,500-19,999	(0.08)	(0.08)	(0.08)
Rural-urban code 7	0.98	0.98	1.00
Not adjacent to metro, 2,500-19,999	(0.09)	(0.09)	(0.09)
Rural-urban code 8	1.01	1.01	0.97
Adjacent to a metro, < 2,500	(0.21)	(0.21)	(0.20)
Rural-urban code 9	0.82	0.81	0.81
Not adjacent to a metro, < 2,500	(0.16)	(0.16)	(0.17)
Observations	~127,000	~127,000	~127,000
State FE	YES	YES	YES
Year FE	YES	YES	YES

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

The results in Table 6 are displayed as an odds ratio. Hence, for example, given a Hispanic-American business owner is female, the business is 1.06 times less likely to continue in business in the next year than a Hispanic-American business owned by a male. The only ethnicity that has a statistically significant impact on the probability of survival in the next year is Puerto Rican. If a Hispanic-American business owner is Puerto Rican, the odds of survival in the next year decrease by 1.19 times. Owner education does not have a statistically significant impact on survival rates. Although business age does not have a statistically significant effect,

the owner age range variables do have significance. Indeed, the omitted owner age category is "less than 25 years old," and every other age category except "older than 64" (which is insignificant) increases the odds of survival in the next year by about 15%.

Both working more than 40 hours per week and having the business be the owner's primary source of income increases the odds of survival. The only start-up capital source that has a statistically significant impact is personal savings and it increases the odds of survival by 6%. For expansion capital source, however, using personal savings actually decreases the odds of survival by 10% and using a bank loan increases the odds of survival by 19%. This relationship between using a bank loan for expansion capital is not entirely causal; banks analyze a business's plan, history, etc., and, in turn, are effective at selecting businesses that are relatively likely to survive. Thus only businesses that are a priori relatively likely to survive receive a bank loan. It may be that the relatively low interest rates of bank loans contribute to the survival of a business, but this coefficient cannot be attributed entirely to this effect. In terms of customer types, both having other businesses and individuals as the primary customers increases the odds of survival. Having state or local governments account for more than 10% of sales also increases the odds of survival, while having the federal government account for more than 10% of sales decreases the odds of survival.

The omitted NAICS 2-digit industry is 11: Agriculture, Forestry, Fishing and Hunting. Although all other (non-suppressed) industries except Management of Companies and Enterprises increase the probability of survival, only (1) Wholesale Trade, (2) Real Estate and Rental and Leasing, (3) Education Services, and (4) Health Care and Social Assistance have statistically significant increase in the probability of survival in the Cox specification. Indeed, LOBs in the educational services and health care and social assistance industries were the most

likely to survive and be significant across all specifications. Finally, note that rurality does not appear to change the probability of survival of a LOB.

Table 7 contains the results of this paper's analysis into factors impacting LOB employment growth. These regressions contain the same variables that were in the LOB survival analysis presented above in Table 6 and the same control variable coefficients as listed above are omitted (available in the appendix) for readability.

Recall, the firm- or establishment–level employment growth rate is defined as follows:

$$(5) \quad g_{it} = \frac{(E_{it} - E_{i,t-\tau})}{X_{it}} \quad where\ X_{it} = \frac{1}{2}(E_{it} + E_{i,t-\tau})$$

E_{it} is employment in year t for establishment i and $E_{i,t-\tau}$ is employment in year $t - \tau$ for establishment i. Columns (1), (2), and (3) of Table 6 present the results with $\tau = 1$ and various level of fixed effects. Columns (4) and (5) present the results with $\tau = 2$ and $\tau = 3$, respectively.

Table 7. Pooled OLS Regression Results: LOB Employment Growth

Business Owner Variables	(1)	(2)	(3)	(4)	(5)
	Employment Growth				
	1-year	1-year	1-year	2-year	3-year
Female	0.02**	0.02**	0.02**	0.01	0.03
	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
Mexican	-0.00	-0.00	-0.00	-0.02	-0.01
	(0.01)	(0.01)	(0.01)	(0.02)	(0.02)
Cuban	0.01	0.01	0.01	0.00	0.02
	(0.01)	(0.01)	(0.01)	(0.02)	(0.03)
Puerto Rican	-0.04**	-0.03**	-0.03*	-0.06**	-0.06
	(0.02)	(0.02)	(0.02)	(0.03)	(0.04)
Owner education: High school	0.02	0.02	0.02	0.05**	0.05
	(0.01)	(0.01)	(0.01)	(0.02)	(0.03)
Some college	0.03***	0.03***	0.03**	0.06***	0.09***
	(0.01)	(0.01)	(0.01)	(0.02)	(0.03)
Technical or trade school	0.03**	0.04**	0.04**	0.06**	0.10**
	(0.02)	(0.01)	(0.02)	(0.03)	(0.04)
Associates degree	0.01	0.01	0.01	0.02	0.05
	(0.02)	(0.02)	(0.02)	(0.03)	(0.04)
Bachelor's degree	0.05***	0.05***	0.05***	0.08***	0.11***
	(0.01)	(0.01)	(0.01)	(0.02)	(0.04)
Graduate degree	0.06***	0.06***	0.06***	0.08***	0.13***
	(0.01)	(0.01)	(0.01)	(0.02)	(0.04)
Business Age	-0.01***	-0.01***	-0.01***	-0.02***	-0.02***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Owner age: 25 to 34	-0.01	-0.01	-0.01	-0.01	-0.01
	(0.02)	(0.02)	(0.02)	(0.04)	(0.06)
35 to 44	-0.06***	-0.06***	-0.06***	-0.07*	-0.10*
	(0.02)	(0.02)	(0.02)	(0.04)	(0.06)
45 to 54	-0.06***	-0.06***	-0.06***	-0.09**	-0.13**
	(0.02)	(0.02)	(0.02)	(0.04)	(0.06)
55 to 64	-0.06***	-0.06***	-0.06***	-0.08**	-0.12**
	(0.02)	(0.02)	(0.02)	(0.04)	(0.06)
Older than 64	-0.02	-0.02	-0.02	-0.04	-0.08
	(0.02)	(0.02)	(0.02)	(0.04)	(0.06)
Hours: 20 to 39 hours	0.02	0.02	0.02	0.03	0.06
	(0.02)	(0.02)	(0.02)	(0.03)	(0.04)
40	0.03*	0.03*	0.03*	0.04	0.04
	(0.02)	(0.01)	(0.01)	(0.03)	(0.04)
41 to 59	0.02	0.02	0.02	0.03	0.03
	(0.01)	(0.01)	(0.01)	(0.02)	(0.04)
More than 59	0.03**	0.03*	0.03**	0.04*	0.06
	(0.01)	(0.01)	(0.01)	(0.03)	(0.04)

Table 7. (cont'd)

Business Owner Variables	(1)	(2)	(3)	(4)	(5)
	Employment Growth				
	1-year	1-year	1-year	2-year	3-year
Primary income	-0.03***	-0.03***	-0.03***	-0.05***	-0.05**
	(0.01)	(0.01)	(0.01)	(0.02)	(0.03)
Provide goods or services	0.01	0.01	0.01**	0.02*	0.03
	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
Management	-0.00	-0.00	-0.00	-0.01	-0.02
	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
Finances	-0.02**	-0.02**	-0.02**	-0.02	-0.02
	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
Multiple establishments	-0.11***	-0.11***	-0.11***	-0.16***	-0.21***
	(0.01)	(0.01)	(0.01)	(0.02)	(0.02)
State-up capital source: savings	0.01	0.01	0.01	0.01	0.02
	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
Assets	0.02*	0.02*	0.02*	0.04**	0.06**
	(0.01)	(0.01)	(0.01)	(0.02)	(0.03)
Credit	-0.01	-0.01	-0.01	-0.02	-0.04
	(0.01)	(0.01)	(0.01)	(0.02)	(0.03)
Government loan	0.02	0.02	0.02	0.00	-0.00
	(0.01)	(0.01)	(0.01)	(0.02)	(0.04)
Bank loan	-0.00	-0.00	-0.00	0.00	0.02
	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
Venture capital	0.00	0.00	0.00	0.01	0.02
	(0.01)	(0.01)	(0.01)	(0.03)	(0.04)
None needed	0.03*	0.03*	0.03*	0.05*	0.07
	(0.02)	(0.02)	(0.02)	(0.03)	(0.04)
Expansion capital source: savings	0.02**	0.02**	0.02**	0.04**	0.05**
	(0.01)	(0.01)	(0.01)	(0.02)	(0.03)
Assets	0.01	0.01	0.01	0.02	0.02
	(0.02)	(0.02)	(0.01)	(0.02)	(0.04)
Credit	0.01	0.01	0.01	0.02	0.04
	(0.01)	(0.01)	(0.01)	(0.02)	(0.03)
Government loan	0.02	0.02	0.02	-0.02	-0.03
	(0.02)	(0.02)	(0.02)	(0.04)	(0.07)
Bank loan	-0.01	-0.01	-0.01	-0.01	-0.01
	(0.01)	(0.01)	(0.01)	(0.02)	(0.03)
Venture capital	-0.01	-0.01	-0.01	0.02	0.06
	(0.03)	(0.03)	(0.03)	(0.05)	(0.07)
None needed	0.01	0.01	0.01	0.00	0.01
	(0.01)	(0.01)	(0.01)	(0.02)	(0.03)

Table 7. (cont'd)

Business and County Variables	(1)	(2)	(3)	(4)	(5)
	Employment Growth				
	1-year	1-year	1-year	2-year	3-year
>10% sales: federal government	-0.01	-0.01	-0.01	-0.04	-0.10**
	(0.01)	(0.01)	(0.01)	(0.03)	(0.04)
State or local government	0.00	0.00	-0.00	0.00	0.01
	(0.01)	(0.01)	(0.01)	(0.02)	(0.03)
Primary customer: other businesses	0.01	0.00	0.00	-0.01	-0.02
	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
Individuals	0.01	0.01	0.01	0.00	0.01
	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
County-level variables:					
% with: high school	-0.15	-0.06	-0.05	-0.03	-0.29
	(0.18)	(0.21)	(0.21)	(0.33)	(0.48)
Some college	0.25	0.10	0.12	-0.25	-0.39
	(0.16)	(0.22)	(0.22)	(0.35)	(0.47)
Associates degree	-0.41	-0.64*	-0.65*	-0.02	0.24
	(0.29)	(0.39)	(0.39)	(0.61)	(0.83)
Bachelor's degree	-0.04	0.00	-0.00	0.12	-0.27
	(0.19)	(0.23)	(0.23)	(0.37)	(0.52)
Graduate degree	-0.14	-0.20	-0.19	-0.25	-0.58
	(0.23)	(0.26)	(0.26)	(0.40)	(0.57)
Per capita income	0.00	0.00	0.00	0.00	0.00*
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Mean household size	-0.00	-0.01	-0.01	-0.03	-0.03
	(0.02)	(0.03)	(0.03)	(0.04)	(0.06)
% of hh headed by single female	0.14	0.22	0.21	-0.00	-0.31
	(0.24)	(0.28)	(0.28)	(0.37)	(0.54)
Unemployment rate	-0.42	-0.42	-0.37	0.22	-0.11
	(0.29)	(0.34)	(0.34)	(0.59)	(0.90)
Labor force participation rate	0.05	0.01	0.00	0.09	0.14
	(0.10)	(0.11)	(0.11)	(0.16)	(0.22)

Table 7. (cont'd)

NAICS Sector Industry	(1)	(2)	(3)	(4)	(5)
	Employment Growth				
	1-year	1-year	1-year	2-year	3-year
21: Mining, Quarrying, and Oil and Gas Extraction			0.14**	0.32***	0.39**
			(0.06)	(0.10)	(0.15)
22: Utilities			0.14	0.30	0.32
			(0.13)	(0.19)	(0.29)
23: Construction			0.08*	0.20**	0.23*
			(0.05)	(0.09)	(0.13)
31-33: Manufacturing			0.08*	0.18**	0.23*
			(0.05)	(0.09)	(0.13)
42: Wholesale Trade			0.09*	0.24***	0.31**
			(0.05)	(0.09)	(0.13)
44-45: Retail Trade			0.08*	0.19**	0.22*
			(0.05)	(0.09)	(0.13)
48-49: Transportation and Warehousing			0.08*	0.19**	0.26*
			(0.05)	(0.09)	(0.14)
51: Information			0.11**	0.19**	0.17
			(0.05)	(0.09)	(0.15)
52: Finance and Insurance			0.09*	0.21**	0.25*
			(0.05)	(0.09)	(0.13)
53: Real Estate, Rental and Leasing			0.10**	0.19**	0.22
			(0.05)	(0.09)	(0.14)
54: Professional, Scientific, and Technical Services			0.07	0.20**	0.23*
			(0.05)	(0.09)	(0.13)
55: Management of Companies and Enterprises			0.09	0.20*	0.25
			(0.06)	(0.11)	(0.16)
56: Administrative Support, Waste Management, Remediation Services			0.07	0.16*	0.18
			(0.05)	(0.09)	(0.14)
61: Educational Services			0.06	0.16*	0.18
			(0.05)	(0.09)	(0.14)
62: Health Care and Social Assistance			0.06	0.18**	0.17
			(0.05)	(0.09)	(0.14)
71: Arts, Entertainment, & Recreation			0.08	0.24**	0.35**
			(0.06)	(0.11)	(0.17)
72: Accommodation & Food Services			0.07	0.19**	0.22*
			(0.05)	(0.09)	(0.13)
81: Other Services			0.05	0.14	0.16
			(0.05)	(0.09)	(0.13)
92: Public Administration			0.51***	0.01	-1.82***
			(0.20)	(0.49)	(0.15)

Table 7. (cont'd)

County Variables	(1)	(2)	(3)	(4)	(5)
	Employment Growth				
	1-year	1-year	1-year	2-year	3-year
Amenities	-0.00	-0.01	-0.01	0.02*	0.03*
	(0.00)	(0.01)	(0.01)	(0.01)	(0.02)
Rural-urban code 2	-0.03***	-0.03**	-0.03**	-0.02	-0.04
Metro area, 0.25 to 1 million pop.	(0.01)	(0.01)	(0.01)	(0.02)	(0.03)
Rural-urban code 3	-0.04***	-0.04***	-0.04***	-0.01	-0.05
Metro area, < 250,000	(0.01)	(0.02)	(0.02)	(0.03)	(0.04)
Rural-urban code 4	-0.02	-0.03	-0.03	-0.00	-0.06
Adjacent to metro, ≥ 20,000	(0.02)	(0.02)	(0.02)	(0.03)	(0.05)
Rural-urban code 5	-0.06***	-0.07***	-0.07***	-0.06	-0.15**
Not adjacent to metro, ≥ 20,000	(0.02)	(0.03)	(0.03)	(0.04)	(0.07)
Rural-urban code 6	-0.00	-0.01	-0.02	-0.03	-0.06
Adjacent to metro, 2,500-19,999	(0.03)	(0.03)	(0.03)	(0.04)	(0.06)
Rural-urban code 7	-0.05	-0.06	-0.06*	-0.08	-0.12
Not adjacent to metro, 2,500-19,999	(0.03)	(0.03)	(0.03)	(0.05)	(0.09)
Rural-urban code 8	0.00	-0.02	-0.02	0.14	0.20
Adjacent to a metro, < 2,500	(0.04)	(0.05)	(0.05)	(0.09)	(0.16)
Rural-urban code 9	-0.02	-0.02	-0.03	-0.01	0.03
Not adjacent to a metro, < 2,500	(0.04)	(0.04)	(0.04)	(0.07)	(0.09)
Constant	0.32*	0.33	0.75***	0.12	-1.58***
	(0.18)	(0.21)	(0.28)	(0.60)	(0.50)
Observations	~112,900	~112,900	~112,900	~86,900	~62,700
R^2	0.03	0.03	0.04	0.05	0.06
Year FE	YES	YES	YES	YES	YES
State FE		YES	YES	YES	YES
NAICS Sector FE			YES	YES	YES

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

As Table 7 shows, the coefficients for 1-year employment growth appear mostly robust to the various levels of fixed effects. As a result, this paper's discussion of the employment growth results focus on columns (3) through (5) of Table 7. The results show that, on average, if a Hispanic-American owner is female, her 1-year employment growth rate is 2 percentage points

higher. This effect becomes statistically insignificant for the 2 and 3-year growth rates, however. If the LOB is owned by a Puerto Rican, 1-year and 2-year employment growth decreases by 3 and 6 percentage points, respectively. The impact of a Puerto Rican owner becomes insignificant on 3-year employment growth. Most county-level factors are statistically insignificant, though rural-urban continuum codes 2, 3, and 5 all have a significant and negative effects on employment growth in multiple specifications. Code 5, defined as an "urban population of 20,000 or more, not adjacent to a metro area" is the most robust to specification. The USDA ERS amenities level has a significant and positive effect on the 2 and 3-year employment growth rates.

All of the owner education levels have a positive and significant coefficient in a least one of the specifications, except associate's degree, which is insignificant in all specifications. As in the other regressions, the omitted category is less than a high school degree, which has an implied negative sign. The significant and positive coefficients on the various education levels increase with education level and as we increase τ. Business age has a significant and negative effect on employment growth, as does owner age. Once again, the effects appear to compound as we increase the time interval of the analysis. The categories of working 40 hours and more than 59 hours a week both have positive association with employment growth.

If the LOB is the owner's primary source of income, there is a statistically significant and negative effect on employment growth. If the owner's main function in the business is to provide the good or the service, the 1 and 2 year employment growth rates increase by 1 and 2 percentage points, respectively. If the owner's that main function in the business is finances, the 1 year employment growth rate decreases by 1 percentage points, though this effect becomes

insignificant in the 2 and 3-year specifications. If a business has multiple establishments, the 1, 2, and 3-year growth rates decrease by 11, 16, and 21 percentage points, respectively.

In terms of the impact of start-up capital sources, using assets reduces the 1, 2, and 3-year employment growth rates by 2, 4, and 6 percentage points, respectively. Not needed start-up capital also increases the employment growth rate. For expansion capital, only personal savings is significant, and it has a statistically significant and positive effect on employment growth in all of the specifications. As in the survival analysis, the omitted 2-digit NAICS category is 11: Agriculture, Forestry, Fishing and Hunting. Each of the following 2-digit NAICS coded industries has a statistically significant effect on employment growth in the 1-year specification: Mining, Quarrying, and Oil and Gas Extraction; Construction; Manufacturing; Wholesale Trade; Retail Trade; Transportation and Warehousing; Information; Finance and Insurance; and Real Estate, Rental and Leasing. Furthermore, this impact appears to affect growth nonlinearly over time, increasing the employment growth rate by a larger amount as the employment growth timespan increases. All of the other included industries except Utilities and Other Services have a statistically significant and positive effect on the employment growth rate in at least one of the specifications.

Conclusions and Suggestions for Future Research

The results on relative associations between owner, business, and geographical characteristics and the owner's ethnicity presented in Table 5 are similar to past results, but there are some important differences. For example, data from the 2005 National Minority Business Owners Survey suggests that in comparison to Korean-Americans, Mexican-American business owners borrow more from family, friends, suppliers, and credit cards and have a lower

proportion of bank loan debt (Haynes, Onochie, and Lee 2008). This paper's results imply this relative association between Korean and Mexican-Americans is more general, and that Asian-American business owners are more likely than Latino-Americans to take on bank loan debt and less likely to take on credit card debt. Further, the results do not support the first hypothesis (H1), that Latino business owners primarily finance their businesses with personal savings. Indeed, only Black business owners are less associated with using personal savings, while White business owners do not differ significantly in their association and Asian business owners are more associated with using personal savings.

The findings of Martinez et al. (2011) contradicted those of previous studies (Haynes, Onochie, and Lee 2008, Cavalluzzo and Wolken 2005, Granier 2006, and Blanchard, Zhao, and Yinger 2008) on the types of startup funds that Latino business owners use, as they did not find that Latino business owners use fewer formal funds compared to Whites. Martinez et al. (2011) did find that African Americans were the only race group significantly less likely to use formal funds compared to Whites. While our results do not separate formal from informal specifically, they do show that LOB are more associated with using credit card and government loan or government guaranteed loans than White business owners, while White business owners are more likely to use bank loans or venture capital. Similar to Martinez et al. (2011), the results show that only Black business owners are less likely than LOB to use bank loans. Hence we find support for H2, that Latino business owners are less likely to finance their business with bank loans. These results are consistent with the findings of Blanchard, Zhao, and Yinger (2008), which show that Latinos receive higher interest rates on bank loans, which also discourages them from seeking formal funds for their businesses.

Using data from the Kauffman Foundation, Martinez and Avila (2011) find that neither Latino nor Latina business owners differ significantly from White men or women in terms of having a college degree. The results presented in Table 5 contradict those findings and show that White business owners have a stronger association with every level of education except "less than a high school degree." Thus we can reject H3, that Latino business owners do not differ significantly from White business owners in their likelihood of having a college degree. Martinez and Avila (2011) also find that Latinos owned significantly fewer home-based businesses compared to White men. While the Census does not have data on home-based businesses, this paper's results show that Latinos are significantly more associated with husband and wife businesses; businesses owned by couples may be associated with home-based business operations.

In general, it is true that Latinos are more likely to work in sectors thought to be relatively vulnerable, such as the services (Puryear et al. 2008, Robles and Cordero-Guzmán 2007), construction, wholesale trade, and retail trade sectors (U.S. Census Bureau 2010). Martinez et al. (2011) explain LOB lack financial capital to enter high-barrier industries, and thus are often associated with low-barrier industries. High barrier industries are those requiring advanced educational degrees or large amounts of startup capital. Low barrier industries include some food services and construction. High barrier industries include professional services and finance or insurance.

Martinez and Avila (2011) find that Latina-owned businesses are a large percentage of businesses in the low-barrier sector and a small percentage of businesses in the high-barrier sectors compared to White-owned businesses, but not that Latino-owned businesses were concentrated in the low-barrier sectors compared to White men-owned businesses, or represented

a significantly smaller percentage of businesses within the high-barrier sectors. These results contradict Bates, Lofstrom, and Servnon (2010). Our results do not separate associations between Latinos and Latinas, but do find that (with our numerous controls) the only sector significantly more associated with Latino/a owners, relative to Whites, is accommodation and food services, which provides partial support for our fourth hypothesis (H4), that LOB are more associated with low-barrier industries such as the services, construction, wholesale trade, and retail trade sectors. Further, although the results do show that White-owned business are more associated with the high-barrier industries of Finance and Insurance, and Professional, Scientific, and Technical Services, LOB either do no differ significantly or are more associated with these high-barrier industries. Thus the results do partially support H5, that LOB are less associated with high-barrier industries such as professional services, and finance or insurance. However, an alternative hypothesis, that White business owners are more associated than other ethnicities with high-barrier industries, may be more accurate.

It may be that the sample size on LOB in the Kauffman Firm Survey is too small. As Martinez et al. (2011) point out, Latinos represent a very small number in the sample ($n = 244$ for primary owners) in the Kauffman Firm Survey data. The SBO data used for this paper's measures of ethnic associations had well over $n = 100{,}000$ for LOB. It is also important to note, however, that the Kauffman sample used in Martinez et al. (2011) is only on LOB start-ups, which may have different factor associations than LOB in general, which the SBO samples.

The results here confirm the results of Fairlie (2005), which found that Latinos' younger age contributes to lower survival rates from business. Indeed, Table 6 shows that Latino business owners are less associated with older ages and Table 7 shows that older Latino business owners are more likely to continue in business. Other results in the survival analysis make substantial

contributions to the literature. Further examining the survival analysis, one of the few statistically significant local population factors in the results is that the percent of the county with some college but no degree as their highest educational attainment decreases the odds of LOB survival in the next year. Given that the model controls for owner education levels, it may be that this coefficient is biased by the unobserved variable "county contains a major university." For example, it is possible that areas lacking a university may cause many people to attempt online degrees, where the non-completion rate is higher than traditional classroom-based programs (Patterson and McFadden 2009). Alternatively, a boom-bust economy might produce a high level of degree non-completions as people alternate between lack of tuition funds and remunerative fulltime non-degree work. As with the boom-bust example, it may also be that areas with many degree non-completers are relatively difficult for business survival for reasons only tangentially related to the local population's education level. Such a conclusion may follow Lichtenstein and Lyons (2006) entrepreneurial pipeline model, in that lack of higher education opportunities may make a county less viable for start-ups because of a lack of larger employers that might serve as customers or role models.

This paper also shows that the results of Robb & Coleman (2009), that women business owners also have more difficulty keeping their businesses in operation compared to male business owners, is applicable to Latinas. Thus we find support for our fifth hypothesis (H5), Latina-owned businesses are less likely to survive than Latino-owned businesses. Bates, Lofstrom, and Servnon (2010) find that businesses within the low barrier industries face a greater likelihood of going out of business compared to businesses within the high barrier industries. The analysis presented here shows that construction and food service do not have low survival rates relative to other industries and thus we can reject our final hypothesis (H7), that LOB in

low-barrier industries are less likely to survive. Unfortunately, the coefficient on the finance and insurance industry is suppressed in the survival analysis, so it is difficult to compare this paper's results to prior work on high-barrier industries.

Past research has also found that a factor impacting Latino business survival and growth is the lack of financial resources, and, as a result, Latinos tend to finance their businesses with personal savings or informal loans from friends, family, or moneylenders (Granier 2006, Haynes, Onochie, and Lee 2008, Raijman and Tienda 2000). In this paper's survival analysis, the only significant coefficient on start-up capital source was on personal savings, but it actually increases the odds of survival of a Latino business by 4%. Indeed this paper supports the slightly different narrative on start-up capital that only Black business owners are less associated with using personal savings than Latinos. It may still be the case, however, that LOB are more likely to use informal loans from friends or family and moneylenders.

Bates, Lofstrom, and Servnon (2010) also suggest business owners with large amounts of personal capital to spend on business startup will be able to open more lucrative businesses in the high-technology, high-barrier industries, while others are more limited in their ability to expand. Our employment growth results do not support such a conclusion. Indeed, LOB in the high-barrier industries of professional services or finance have similar or lower employment growth rates than LOB in the low-barrier industries of wholesale or retail trade. Further, this paper shows that LOB using savings for start-up capital does not have a statistically significant impact on employment growth, and using personal assets for start-up or expansion capital actually increases the employment growth. Recall also that this paper shows LOB as more associated with the $5,000 to $10,000 range of start-up capital than the other ethnicities. Further, LOB are more likely to be associated with the ranges in $10,000 to $100,000 than White and Black-

Americans, but less likely than Asian-Americans (though the $50,000 to $100,000 range is suppressed for Asian-Americans). The LOB are less associated with all of the ranges over $100,000 than Asian and White-Americans, but more associated with those (non-suppressed) ranges than Black-Americans. Hence there is indeed evidence that LOB are relatively likely to start a business with a small amount of non-personal capital, but this paper does not support the conclusion that this small amount of start-up capital, in turn, decreases their chances of survival or limits their future growth.

Martinez and Avil's (2011) Cox survival analysis suggest education is significant to business survival controlling for personal startup funds. Though this paper did not find a statistically significant impact of any education level on LOB survival, it did show that all owner education levels except associate's degree have a positive and significant coefficient on employment growth. The implied sign on the omitted category (less than a high school degree), is negative. Furthermore, this relationship appears to be nonlinear, with an increasing effect on employment growth rate as the time horizon increases.

Finally, an advantage of access to such a large dataset is the ability to examine specific Latino subgroups. Unfortunately, the only subgroup detail available in the SBO was on whether the Latino business owner has Mexican, Cuban, or Puerto Rican ancestry. The only ancestral origin that was significant in either the survival or employment growth regression was Puerto Rican origin. Note that this paper uses data on the continental 48 states and Washington D.C., and excludes businesses in Puerto Rico. On average, having Puerto Rican ancestry decreases the odds of business survival by 19% and lowers the employment growth rate by 3 percentage points over 1 year and 6 percentage points over 2 years. These results help provide empirical evidence to support recent discussions of the low barriers to entry in the United States for Puerto Rican

migrants and the accompanying struggles (e.g. Abel and Deitz 2015, *Economist* 2015), and is consistent with evidence that out-migrants from Puerto Rico are relatively low-skill (Borjas 2007).

Puerto Rican migrants' experience in the continental United States is significantly different than that of Cuban immigrants.[11] Cuban immigration to the United States increased substantially after the overthrow of the Fulgencio Batista regime by Castro-led revolution in 1959, but immigration continued through the 1990s. It is still true that compared with the overall immigrant and U.S.-born populations, Cubans are less likely to be proficient in English, have lower educational attainment, and earn a lower household income (Rusin, Zong, and Batalova 2015). It is also true, however, that Cuban immigrants, are older, have a higher level of education, and higher median household income compared with the rest of the Hispanic population in the United States (*Pew Research* 2006). Despite these differences, the impact of the LOB being Cuban-owned is statistically insignificant.

Future research with this or other data may want to more closely examine the cause of businesses exiting the data; while most exits are likely the result of failure, some may result from buy-out or a marketing change. Another limitation of this data results from Latino business owners tending to participate in smaller informal economic activities, such as street vending businesses and, consequently, formal sources of identification would not capture the full range of self-employment opportunities in which Latinos participate (Martinez, et al. 2011). Indeed, Raijman (2001) suggests census data do not adequately cover some types of economic activities, such as part-time and irregular work or informal self-employment. Future research may want to examine this aspect of Latino business ownership.

[11] The use of the word "migrants" to refer to individuals from Puerto Rico rather than "immigrants" is purposeful as Puerto Rico is part of the United States.

Another area for future research is using big data to examine how various factors impact Latino relative to Latina survival rates. Both Martinez and Avil (2011) and Zuiker et al. (2003) found significant gender interactions with various factors impacting survival, but as mentioned above, many public datasets contain relatively few observations on Latinos and Latinas. Further, as Latinos are a heterogeneous group, future research may also want to examine the relative associations of various Latino origins with business, owner, and geographical characteristics, as this paper did with Latino, Asian, Black, and White business owners in Table 5. These areas for future research highlight how this research provides new insights into an important growth sector in the U.S. economy. Moving forward, federal, state, and local policy makers interested in economic development can use these results to not only guide their future investigations into disparities among ethnic groups in business ownership, but also increase the accuracy and thereby impact of their economic development programs.

APPENDIX

Table A1. Multinomial Logit Results (Omitted Relative Risk Ratio Coefficients)

Business and County Variables	(1) Asian-American	(2) Black-American	(3) White-American
Business type: coop	4.68***	0.89	0.82
	(1.08)	(0.24)	(0.18)
Estate	1.03	XX	XX
	(0.51)	XX	XX
Nonprofit	2.18**	8.20***	1.31
	(0.85)	(2.01)	(0.28)
Public	0.97	0.97	0.56***
	(0.06)	(0.06)	(0.03)
Husband and wife	XX	0.90***	0.95***
	XX	(0.02)	(0.02)
Establ. employment	0.99***	1.00*	1.00***
	(0.00)	(0.00)	(0.00)
Establishment payroll	1.00**	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
Firm payroll	1.00*	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
Firm employment	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
County-level variables:			
Population	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
Population squared	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
Population density	1.00***	1.00**	1.00***
	(0.00)	(0.00)	(0.00)
25 to 34 population	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
35 to 44 population	1.00	1.00**	1.00
	(0.00)	(0.00)	(0.00)
45 to 54 population	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
55 to 64 population	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
> 64 population	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)

Table A1. (cont'd)

County Variables	(1) Asian-American	(2) Black-American	(3) White-American
Female population squared	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
Black population	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
Black population squared	1.00	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
Asian population	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
Asian population squared	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
Hispanic population	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
Hispanic population squared	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
% with: High school	947.09***	16,878.84***	13,089.72***
	(416.08)	(8,357.78)	(4,205.45)
Some college	39.80***	1,210.90***	8.59***
	(19.06)	(622.52)	(3.06)
Associates degree	188.23***	107.75***	312.14***
	(160.48)	(100.93)	(199.73)
Bachelor's degree	72.50***	580.62***	160.48***
	(31.81)	(291.45)	(53.78)
Graduate degree	3.80**	3.38**	0.83
	(2.03)	(2.03)	(0.33)
Build permits	1.00	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
Industry employment	1.00	1.00	1.00***
	(0.00)	(0.00)	(0.00)
State industry employment	1.00***	1.00**	1.00
	(0.00)	(0.00)	(0.00)

Table A1. (cont'd)

State Variables	(1) Asian-American	(2) Black-American	(3) White-American
Arkansas	0.60***	0.49***	0.67***
	(0.09)	(0.07)	(0.09)
California	0.57***	XX	0.35***
	(0.07)	XX	(0.04)
Colorado	0.29***	0.10***	0.23***
	(0.03)	(0.01)	(0.02)
Connecticut	0.49***	0.17***	0.38***
	(0.06)	(0.02)	(0.04)
Delaware	0.85	0.31***	0.39***
	(0.15)	(0.06)	(0.06)
District of Columbia	0.44***	0.28***	0.21***
	(0.06)	(0.04)	(0.03)
Florida	0.27***	0.12***	0.15***
	(0.03)	(0.01)	(0.01)
Georgia	0.74***	0.52***	0.45***
	(0.08)	(0.05)	(0.04)
Idaho	XX	0.05***	0.49***
	XX	(0.01)	(0.06)
Illinois	0.40***	XX	0.29***
	(0.05)	XX	(0.03)
Indiana	XX	XX	XX
	XX	XX	XX
Iowa	0.37***	0.07***	0.37***
	(0.06)	(0.01)	(0.05)
Kansas	0.32***	0.14***	0.28***
	(0.04)	(0.02)	(0.03)
Kentucky	1.13	0.56***	1.18
	(0.17)	(0.08)	(0.15)
Louisiana	0.65***	0.44***	0.56***
	(0.08)	(0.05)	(0.06)
Maine	XX	XX	XX
	XX	XX	XX
Maryland	0.68***	0.41***	0.37***
	(0.08)	(0.04)	(0.04)
Massachusetts	0.59***	0.14***	0.47***
	(0.07)	(0.02)	(0.05)
Michigan	XX	0.23***	0.53***
	XX	(0.03)	(0.05)
Minnesota	0.52***	XX	0.60***
	(0.07)	XX	(0.07)

Table A1. (cont'd)

State Variables	(1) Asian-American	(2) Black-American	(3) White-American
Missouri	0.81	0.41***	0.80*
	(0.11)	(0.05)	(0.09)
Montana	0.32***	0.05***	0.58***
	(0.08)	(0.02)	(0.10)
Nebraska	0.29***	0.09***	0.31***
	(0.05)	(0.02)	(0.04)
Nevada	XX	XX	0.16***
	XX	XX	(0.02)
New Hampshire	0.60***	XX	0.57***
	(0.12)	XX	(0.10)
New Jersey	0.50***	0.10***	0.21***
	(0.06)	(0.01)	(0.02)
New Mexico	0.09***	0.02***	0.07***
	(0.01)	(0.00)	(0.01)
New York	0.42***	0.12***	0.33***
	(0.05)	(0.01)	(0.03)
North Carolina	0.71***	0.71***	0.58***
	(0.08)	(0.08)	(0.06)
North Dakota	XX	XX	XX
	XX	XX	XX
Ohio	0.76**	0.28***	0.63***
	(0.09)	(0.03)	(0.07)
Oklahoma	0.56***	0.19***	0.43***
	(0.07)	(0.02)	(0.05)
Oregon	0.80*	0.08***	0.48***
	(0.10)	(0.01)	(0.05)
Pennsylvania	0.67***	0.18***	0.44***
	(0.08)	(0.02)	(0.05)
Rhode Island	0.54***	0.19***	0.70***
	(0.09)	(0.04)	(0.10)
South Carolina	0.79*	0.99	0.75**
	(0.10)	(0.13)	(0.09)
South Dakota	XX	XX	XX
	XX	XX	XX
Tennessee	0.79*	0.41***	0.75***
	(0.10)	(0.05)	(0.08)
Texas	0.28***	0.10***	0.18***
	(0.03)	(0.01)	(0.02)
Utah	XX	XX	0.60***
	XX	XX	(0.07)

Table A1. (cont'd)

State and Year Variables	(1) Asian-American	(2) Black-American	(3) White-American
Vermont	XX	XX	XX
	XX	XX	XX
Virginia	0.70***	0.43***	0.38***
	(0.08)	(0.05)	(0.04)
Washington	0.60***	0.11***	0.39***
	(0.07)	(0.01)	(0.04)
West Virginia	1.18	0.22***	0.69*
	(0.25)	(0.05)	(0.13)
Wisconsin	0.39***	XX	0.40***
	(0.05)	XX	(0.05)
Wyoming	0.23***	XX	0.22***
	(0.05)	XX	(0.04)
2003	0.92***	1.12***	0.88***
	(0.01)	(0.01)	(0.01)
2004	0.87***	1.25***	0.79***
	(0.01)	(0.02)	(0.01)
2005	0.84***	1.40***	0.71***
	(0.02)	(0.03)	(0.01)
2006	0.80***	1.43***	0.64***
	(0.02)	(0.04)	(0.01)
2007	0.85***	1.42***	0.66***
	(0.02)	(0.04)	(0.01)

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table A2. Multinomial Logit Results (Logit Coefficients)

Business Variables	(1) Asian-American	(2) Black-American	(3) White-American
Female	-0.08***	0.21***	-0.28***
	(0.02)	(0.02)	(0.01)
Education: high school	0.51***	0.82***	1.20***
	(0.03)	(0.04)	(0.02)
Some college	XX	1.30***	1.56***
	XX	(0.04)	(0.02)
Technical or trade school	0.25***	1.25***	1.32***
	(0.04)	(0.04)	(0.03)
Associate's degree	0.77***	XX	1.53***
	(0.04)	XX	(0.03)
Bachelor's degree	1.54***	1.33***	2.09***
	(0.03)	(0.04)	(0.02)
Graduate degree	1.78***	1.21***	2.02***
	(0.04)	(0.04)	(0.03)
Owner age: 25 to 34	XX	XX	-2.22***
	XX	XX	(0.04)
35 to 44	-1.35***	-0.99***	-2.06***
	(0.05)	(0.05)	(0.04)
45 to 54	-1.25***	-0.72***	-1.65***
	(0.05)	(0.05)	(0.04)
55 to 64	-1.20***	-0.47***	-1.31***
	(0.05)	(0.05)	(0.04)
Older than 64	-1.43***	-0.32***	-0.99***
	(0.05)	(0.06)	(0.04)
Hours: 20 to 39	-0.25***	XX	-0.25***
	(0.03)	XX	(0.02)
40	-0.02	-0.21***	-0.40***
	(0.03)	(0.03)	(0.02)
41 to 59	-0.13***	-0.12***	-0.25***
	(0.03)	(0.03)	(0.02)
More than59	0.04	0.24***	-0.33***
	(0.03)	(0.03)	(0.02)
Primary income	0.15***	-0.53***	-0.04**
	(0.02)	(0.02)	(0.02)
Provide	-0.24***	XX	-0.06***
	(0.02)	XX	(0.01)
Manage	0.09***	0.16***	0.12***
	(0.02)	(0.02)	(0.02)
Financial	0.05***	-0.08***	0.39***
	(0.02)	(0.02)	(0.02)

Table A2. (cont'd)

Business Variables	(1) Asian-American	(2) Black-American	(3) White-American
Business type: Coop	1.54***	-0.12	-0.20
	(0.23)	(0.27)	(0.22)
Estate	0.03	XX	XX
	(0.50)	XX	XX
Nonprofit	0.78**	2.10***	0.27
	(0.39)	(0.25)	(0.21)
Public	-0.03	-0.03	-0.58***
	(0.06)	(0.07)	(0.05)
Husband and wife	XX	-0.11***	-0.06***
	XX	(0.03)	(0.02)
Establishment employment	-0.01***	0.00*	-0.00***
	(0.00)	(0.00)	(0.00)
Establishment payroll	0.00**	-0.00***	0.00***
	(0.00)	(0.00)	(0.00)
Firm payroll	-0.00*	-0.00***	-0.00***
	(0.00)	(0.00)	(0.00)
Firm employment	0.00***	0.00***	0.00***
	(0.00)	(0.00)	(0.00)
Start-up capital amount:	-0.24***	-0.07*	-0.27***
$5,000-$10,000	(0.03)	(0.03)	(0.02)
$10,000-$25,000	0.14***	-0.20***	-0.19***
	(0.03)	(0.04)	(0.02)
$25,000-$50,000	0.28***	-0.23***	-0.14***
	(0.04)	(0.05)	(0.03)
$50,000-$100,000	XX	-0.32***	-0.04
	XX	(0.05)	(0.03)
$100,000-$250,000	0.61***	-0.29***	0.07**
	(0.04)	(0.05)	(0.03)
$250,000-$1 million	0.61***	-0.31***	0.21***
	(0.05)	(0.07)	(0.05)
Greater than $1 million	0.75***	XX	0.37***
	(0.08)	XX	(0.07)

Table A2. (cont'd)

Business Variables	(1) Asian-American	(2) Black-American	(3) White-American
Credit	-0.09***	0.10***	-0.16***
	(0.03)	(0.03)	(0.02)
Government	XX	0.28***	-0.12**
	XX	(0.07)	(0.05)
Bank loan	0.26***	-0.11***	0.17***
	(0.03)	(0.03)	(0.02)
Venture	0.29***	0.06	0.28***
	(0.08)	(0.09)	(0.06)
None needed	-0.20***	-0.12***	-0.08***
	(0.03)	(0.03)	(0.02)
Expansion capital source:	0.11***	0.05**	-0.02
Savings	(0.02)	(0.03)	(0.02)
Credit	-0.31***	-0.07**	0.08***
	(0.03)	(0.03)	(0.02)
Government	-0.21***	0.13	-0.24***
	(0.08)	(0.08)	(0.06)
Bank Loan	-0.12***	-0.32***	0.11***
	(0.03)	(0.03)	(0.02)
Venture	-0.07	XX	-0.08
	(0.10)	XX	(0.08)
None needed	-0.02	-0.18***	0.19***
	(0.02)	(0.02)	(0.02)
>10% Federal	0.09*	0.07	-0.42***
	(0.05)	(0.05)	(0.04)
State or local	-0.40***	0.34***	-0.12***
	(0.04)	(0.04)	(0.03)
Other bususnesses	-0.10***	0.11***	-0.12***
	(0.01)	(0.01)	(0.01)
Individuals	0.02*	-0.11***	0.04***
	(0.01)	(0.01)	(0.01)

Table A2. (cont'd)

County Variables	(1) Asian-American	(2) Black-American	(3) White-American
Population	-0.00***	-0.00***	0.00***
	(0.00)	(0.00)	(0.00)
Population squared	0.00***	0.00**	0.00***
	(0.00)	(0.00)	(0.00)
Population density	-0.00***	-0.00***	-0.00***
	(0.00)	(0.00)	(0.00)
25 to 34 population	0.00	0.00**	0.00
	(0.00)	(0.00)	(0.00)
35 to 44 population	-0.00***	-0.00***	-0.00***
	(0.00)	(0.00)	(0.00)
45 to 54 population	0.00***	0.00***	0.00***
	(0.00)	(0.00)	(0.00)
55 to 64 population	-0.00***	-0.00***	-0.00***
	(0.00)	(0.00)	(0.00)
Female population	-0.00***	-0.00***	-0.00***
	(0.00)	(0.00)	(0.00)
Female population squared	0.00***	0.00***	-0.00***
	(0.00)	(0.00)	(0.00)
Black population	0.00***	0.00***	-0.00***
	(0.00)	(0.00)	(0.00)
Black population squared	0.00	-0.00***	0.00***
	(0.00)	(0.00)	(0.00)
Asian population	0.00***	0.00***	-0.00***
	(0.00)	(0.00)	(0.00)
Asian population squared	-0.00***	-0.00***	0.00***
	(0.00)	(0.00)	(0.00)
Hispanic population	-0.00***	-0.00***	-0.00***
	(0.00)	(0.00)	(0.00)
Hispanic population squared	0.00***	0.00***	0.00***
	(0.00)	(0.00)	(0.00)
% with: high school	6.85***	9.73***	9.48***
	(0.44)	(0.50)	(0.32)
Some college	3.68***	7.10***	2.15***
	(0.48)	(0.51)	(0.36)
Associates degree	5.24***	4.68***	5.74***
	(0.85)	(0.94)	(0.64)
Bachelor's degree	4.28***	6.36***	5.08***
	(0.44)	(0.50)	(0.34)
Graduate degree	1.34**	1.22**	-0.18
	(0.54)	(0.60)	(0.40)

Table A2. (cont'd)

County Variables	(1) Asian-American	(2) Black-American	(3) White-American
Per capita income	0.00***	0.00***	0.00***
	(0.00)	(0.00)	(0.00)
Mean household size	-0.51***	-0.41***	-1.53***
	(0.06)	(0.07)	(0.04)
% headed by single female	1.52***	10.23***	-4.26***
	(0.43)	(0.47)	(0.33)
Unemployment rate	2.87***	0.71	0.49
	(0.82)	(0.94)	(0.63)
Labor force participation rate	0.43*	-0.39	1.22***
	(0.23)	(0.25)	(0.17)
Build permits	0.00	-0.00***	-0.00***
	(0.00)	(0.00)	(0.00)
Industry employment	-0.00	0.00	0.00***
	(0.00)	(0.00)	(0.00)
State industry employment	0.00***	0.00**	-0.00
	(0.00)	(0.00)	(0.00)

Table A2. (cont'd)

NAICS Sector Industry	(1) Asian-American	(2) Black-American	(3) White-American
11: Ag., Forestry, Fishing, Hunting	-0.42*	-0.53**	0.62***
	(0.22)	(0.21)	(0.16)
21: Mining, Quarrying, Oil, and Gas Extraction	-0.69***	-0.59**	1.33***
	(0.25)	(0.28)	(0.19)
22: Utilities	-0.02	0.24	0.73***
	(0.34)	(0.32)	(0.22)
23: Construction	-0.99***	-0.55***	0.36**
	(0.18)	(0.18)	(0.14)
31-33: Manufacturing	0.08	-0.86***	0.45***
	(0.18)	(0.18)	(0.14)
42: Wholesale Trade	0.67***	-0.92***	0.50***
	(0.18)	(0.18)	(0.14)
44-45: Retail Trade	0.68***	-0.47***	0.32**
	(0.17)	(0.18)	(0.14)
48-49: Transportation and Warehousing	-0.31*	XX	-0.15
	(0.18)	XX	(0.14)
51: Information	-0.09	0.16	0.67***
	(0.18)	(0.19)	(0.15)
52: Finance and Insurance	-0.23	-0.12	0.56***
	(0.18)	(0.18)	(0.14)
53: Real Estate, Rental and Leasing	0.22	-0.32*	XX
	(0.17)	(0.18)	XX
54: Professional, Scientific, Technical Services	-0.02	-0.16	0.50***
	(0.17)	(0.18)	(0.14)
55: Management of Companies and Enterprises	0.50**	0.13	0.51**
	(0.24)	(0.26)	(0.20)
56: Administrative Support, Waste Management, Remediation Services	-0.76***	-0.18	-0.13
	(0.18)	(0.18)	(0.14)
61: Educational Services	-0.03	0.00	0.32**
	(0.18)	(0.19)	(0.15)
62: Health Care, Social Assistance	0.09	0.34*	-0.04
	(0.18)	(0.18)	(0.14)
71: Arts, Entertainment, Recreation	-0.47***	0.38**	0.73***
	(0.18)	(0.18)	(0.14)
72: Accommodation, Food Services	1.12***	-0.83***	-0.34**
	(0.18)	(0.18)	(0.14)
81: Other Services	XX	0.01	0.13
	XX	(0.18)	(0.14)

Table A2. (cont'd)

County Variables	(1) Asian-American	(2) Black-American	(3) White-American
Amenities	-0.18***	-0.28***	-0.15***
	(0.02)	(0.02)	(0.01)
Rural-urban code 2	0.01	0.07**	0.12***
Metro area, 0.25 to 1 million pop.	(0.03)	(0.03)	(0.02)
Rural-urban code 3	-0.10**	0.05	0.24***
Metro area, < 250,000	(0.04)	(0.05)	(0.03)
Rural-urban code 4	-0.12**	0.10	0.31***
Adjacent to metro, ≥ 20,000	(0.06)	(0.06)	(0.04)
Rural-urban code 5	-0.12	0.04	0.46***
Not adjacent to metro, ≥ 20,000	(0.08)	(0.09)	(0.06)
Rural-urban code 6	-0.12*	0.20***	0.57***
Adjacent to metro, 2,500-19,999	(0.07)	(0.07)	(0.05)
Rural-urban code 7	-0.29***	-0.18**	0.43***
Not adjacent to metro, 2,500-19,999	(0.08)	(0.09)	(0.06)
Rural-urban code 8	-0.30*	0.47***	0.66***
Adjacent to a metro, < 2,500	(0.17)	(0.16)	(0.13)
Rural-urban code 9	-0.99***	0.11	0.65***
Not adjacent to a metro, < 2,500	(0.17)	(0.16)	(0.11)

Table A2. (cont'd)

State Variables	(1) Asian-American	(2) Black-American	(3) White-American
Arizona	XX	-2.84***	-1.72***
	XX	(0.15)	(0.10)
Arkansas	-0.51***	-0.71***	-0.41***
	(0.15)	(0.15)	(0.13)
California	-0.57***	XX	-1.06***
	(0.13)	XX	(0.11)
Colorado	-1.24***	-2.25***	-1.49***
	(0.12)	(0.13)	(0.10)
Connecticut	-0.72***	-1.78***	-0.96***
	(0.12)	(0.13)	(0.11)
Delaware	-0.17	-1.16***	-0.94***
	(0.18)	(0.18)	(0.16)
District of Columbia	-0.83***	-1.27***	-1.58***
	(0.15)	(0.15)	(0.13)
Florida	-1.31***	-2.11***	-1.93***
	(0.11)	(0.11)	(0.09)
Georgia	-0.30***	-0.66***	-0.79***
	(0.11)	(0.11)	(0.09)
Idaho	XX	-3.10***	-0.72***
	XX	(0.32)	(0.12)
Illinois	-0.92***	XX	-1.25***
	(0.12)	XX	(0.10)
Indiana	XX	XX	XX
	XX	XX	XX
Iowa	-0.99***	-2.62***	-1.00***
	(0.16)	(0.19)	(0.14)
Kansas	-1.14***	-1.95***	-1.28***
	(0.13)	(0.14)	(0.11)
Kentucky	0.12	-0.58***	0.17
	(0.15)	(0.15)	(0.13)
Louisiana	-0.44***	-0.83***	-0.58***
	(0.12)	(0.12)	(0.11)
Maine	XX	XX	XX
	XX	XX	XX
Maryland	-0.38***	-0.90***	-0.99***
	(0.11)	(0.11)	(0.10)
Massachusetts	-0.53***	-1.96***	-0.75***
	(0.12)	(0.13)	(0.10)
Michigan	XX	-1.49***	-0.64***
	XX	(0.12)	(0.10)

Table A2. (cont'd)

State Variables	(1) Asian-American	(2) Black-American	(3) White-American
Minnesota	-0.65***	XX	-0.51***
	(0.14)	XX	(0.12)
Mississippi	0.55***	0.86***	0.66***
	(0.17)	(0.17)	(0.16)
Missouri	-0.21	-0.89***	-0.22*
	(0.13)	(0.13)	(0.11)
Montana	-1.14***	-2.95***	-0.54***
	(0.24)	(0.35)	(0.17)
Nebraska	-1.25***	-2.37***	-1.16***
	(0.17)	(0.19)	(0.13)
Nevada	XX	XX	-1.86***
	XX	XX	(0.11)
New Hampshire	-0.51***	XX	-0.56***
	(0.19)	XX	(0.18)
New Jersey	-0.68***	-2.28***	-1.56***
	(0.11)	(0.11)	(0.09)
New Mexico	-2.40***	-3.84***	-2.70***
	(0.13)	(0.18)	(0.10)
New York	-0.88***	-2.15***	-1.09***
	(0.12)	(0.12)	(0.10)
North Carolina	-0.34***	-0.35***	-0.55***
	(0.11)	(0.11)	(0.10)
North Dakota	XX	XX	XX
	XX	XX	XX
Ohio	-0.27**	-1.29***	-0.47***
	(0.12)	(0.12)	(0.11)
Oklahoma	-0.58***	-1.67***	-0.84***
	(0.13)	(0.13)	(0.11)
Oregon	-0.23*	-2.48***	-0.73***
	(0.12)	(0.15)	(0.11)
Pennsylvania	-0.40***	-1.70***	-0.82***
	(0.12)	(0.12)	(0.11)
Rhode Island	-0.62***	-1.65***	-0.36***
	(0.16)	(0.19)	(0.14)
South Carolina	-0.24*	-0.01	-0.29**
	(0.13)	(0.13)	(0.11)
South Dakota	XX	XX	XX
	XX	XX	XX
Tennessee	-0.24*	-0.89***	-0.29***
	(0.13)	(0.12)	(0.11)

Table A2. (cont'd)

State and Year Variables	(1) Asian-American	(2) Black-American	(3) White-American
Texas	-1.27***	-2.31***	-1.69***
	(0.11)	(0.11)	(0.09)
Utah	XX	XX	-0.51***
	XX	XX	(0.12)
Vermont	XX	XX	XX
	XX	XX	XX
Virginia	-0.35***	-0.84***	-0.97***
	(0.11)	(0.11)	(0.09)
Washington	-0.51***	-2.25***	-0.95***
	(0.12)	(0.13)	(0.10)
West Virginia	0.17	-1.51***	-0.37*
	(0.21)	(0.25)	(0.19)
Wisconsin	-0.94***	XX	-0.90***
	(0.13)	XX	(0.11)
Wyoming	-1.48***	XX	-1.50***
	(0.21)	XX	(0.17)
2003	-0.09***	0.12***	-0.12***
	(0.01)	(0.01)	(0.01)
2004	-0.14***	0.22***	-0.23***
	(0.01)	(0.02)	(0.01)
2005	-0.17***	0.34***	-0.34***
	(0.02)	(0.02)	(0.02)
2006	-0.23***	0.36***	-0.44***
	(0.02)	(0.03)	(0.02)
2007	-0.17***	0.35***	-0.42***
	(0.03)	(0.03)	(0.02)
Constant	-2.26***	-5.28***	4.61***
	(0.48)	(0.54)	(0.36)
Pseudo R^2	0.224	0.224	0.224
Observations	~11,847,500	~11,847,500	~11,847,500

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

Table A3. Cox Proportional Hazard, Exponential, Weibull Survival Model Results (Omitted Odds Ratio Coefficients)

Business Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Business type: Husband and wife	1.43***	1.43***	1.44***
	(0.10)	(0.10)	(0.10)
Estate	XX	XX	XX
	XX	XX	XX
Coop	0.92	0.92	0.92
	(0.18)	(0.18)	(0.19)
Nonprofit	1.62**	1.62**	1.68**
	(0.33)	(0.33)	(0.37)
Public	XX	XX	XX
	XX	XX	XX
Establishment employment	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
Establishment payroll	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
Firm employment	1.00**	1.00**	1.00**
	(0.00)	(0.00)	(0.00)
Firm payroll	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)
Hispanic establishments	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Mexican establishments	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Puerto Rican establishments	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Cuban establishments	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Hispanic employment	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Mexican employment	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Puerto Rican employment	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Cuban employment	1.00***	1.00***	1.00***
	(0.00)	(0.00)	(0.00)

Table A3. (cont'd)

County Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Interaction: Mexican & Mexican	1.00	1.00	1.00
population	(0.00)	(0.00)	(0.00)
Puerto Rican & Puerto Rican	1.00**	1.00**	1.00**
population	(0.00)	(0.00)	(0.00)
Cuban & Cuban population	1.00**	1.00**	1.00***
	(0.00)	(0.00)	(0.00)
Mexican & Mexican establishments	1.00*	1.00*	1.00**
	(0.00)	(0.00)	(0.00)
Puerto Rican & Puerto Rican	1.00	1.00	1.00
establishments	(0.00)	(0.00)	(0.00)
Cuban & Cuban establishments	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Mexican & Mexican employment	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Puerto Rican & Puerto Rican	1.00	1.00	1.00
employment	(0.00)	(0.00)	(0.00)
Cuban & Cuban employment	1.00	1.00	1.00*
	(0.00)	(0.00)	(0.00)
Population	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Population squared	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Population density	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
25 to 34 population	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
35 to 44 population	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
45 to 54 population	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
55 to 64 population	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
> 64 population	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)

Table A3. (cont'd)

County Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Female population	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Female population squared	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Black population	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Black population squared	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Asian population	1.00*	1.00*	1.00*
	(0.00)	(0.00)	(0.00)
Asian population squared	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Hispanic population	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Hispanic population squared	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Mexican population	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Mexican population squared	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Puerto Rican population	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Puerto Rican population squared	1.00***	1.00***	1.00**
	(0.00)	(0.00)	(0.00)
Cuban population	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Cuban population squared	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Mean household size	1.01	1.01	1.02
	(0.09)	(0.09)	(0.09)
% of hh headed by single female	1.27	1.27	1.21
	(0.78)	(0.78)	(0.76)
Build permits	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
Industry employment	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)
State industry employment	1.00	1.00	1.00
	(0.00)	(0.00)	(0.00)

Table A3. (cont'd)

State Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Arizona	0.88	0.89	0.91
	(0.13)	(0.13)	(0.14)
Arkansas	0.84	0.86	0.89
	(0.18)	(0.19)	(0.20)
California	0.91	0.91	0.92
	(0.15)	(0.16)	(0.16)
Colorado	1.03	1.05	1.08
	(0.15)	(0.15)	(0.16)
Connecticut	1.12	1.14	1.16
	(0.20)	(0.20)	(0.21)
Delaware	XX	XX	XX
	XX	XX	XX
District of Columbia	1.08	1.11	1.19
	(0.20)	(0.21)	(0.23)
Florida	1.03	1.04	1.08
	(0.15)	(0.15)	(0.16)
Georgia	0.94	0.95	0.97
	(0.13)	(0.13)	(0.14)
Idaho	0.76	0.76	0.74
	(0.15)	(0.15)	(0.15)
Illinois	0.83	0.85	0.90
	(0.13)	(0.13)	(0.15)
Indiana	0.87	0.88	0.91
	(0.15)	(0.15)	(0.16)
Iowa	0.95	0.97	1.00
	(0.18)	(0.19)	(0.20)
Kansas	0.89	0.90	0.93
	(0.16)	(0.16)	(0.17)
Kentucky	0.99	0.99	0.98
	(0.19)	(0.19)	(0.19)
Louisiana	1.04	1.04	1.03
	(0.16)	(0.16)	(0.16)
Maine	XX	XX	XX
	XX	XX	XX
Maryland	0.74**	0.75*	0.78
	(0.11)	(0.11)	(0.12)
Massachusetts	1.08	1.09	1.10
	(0.17)	(0.17)	(0.18)
Michigan	1.02	1.03	1.05
	(0.15)	(0.16)	(0.16)

Table A3. (cont'd)

State Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Minnesota	1.12	1.14	1.20
	(0.19)	(0.20)	(0.21)
Mississippi	1.27	1.28	1.32
	(0.26)	(0.26)	(0.28)
Missouri	0.89	0.90	0.94
	(0.15)	(0.15)	(0.16)
Montana	XX	XX	XX
	XX	XX	XX
Nebraska	1.06	1.07	1.09
	(0.20)	(0.21)	(0.22)
Nevada	0.90	0.91	0.94
	(0.16)	(0.16)	(0.17)
New Hampshire	0.71	0.72	0.74
	(0.20)	(0.20)	(0.21)
New Jersey	XX	XX	XX
	XX	XX	XX
New Mexico	0.81	0.82	0.83
	(0.12)	(0.12)	(0.12)
New York	XX	XX	XX
	XX	XX	XX
North Carolina	0.90	0.91	0.92
	(0.13)	(0.13)	(0.14)
North Dakota	XX	XX	XX
	XX	XX	XX
Ohio	0.89	0.91	0.93
	(0.14)	(0.14)	(0.15)
Oklahoma	0.95	0.97	0.98
	(0.15)	(0.15)	(0.16)
Oregon	0.77	0.78	0.79
	(0.13)	(0.13)	(0.14)
Pennsylvania	0.93	0.94	0.96
	(0.15)	(0.15)	(0.16)
Rhode Island	1.06	1.09	1.13
	(0.24)	(0.25)	(0.27)
South Carolina	0.81	0.82	0.85
	(0.15)	(0.15)	(0.16)
South Dakota	1.10	1.13	1.20
	(0.28)	(0.29)	(0.31)
Tennessee	0.86	0.86	0.86
	(0.14)	(0.14)	(0.15)

Table A3. (cont'd)

State and Year Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Texas	1.00	1.00	1.00
	(0.14)	(0.14)	(0.15)
Utah	0.89	0.90	0.92
	(0.18)	(0.18)	(0.19)
Vermont	XX	XX	XX
	XX	XX	XX
Virginia	0.97	0.98	1.00
	(0.14)	(0.14)	(0.15)
Washington	0.95	0.96	0.97
	(0.14)	(0.15)	(0.15)
West Virginia	1.08	1.08	1.08
	(0.24)	(0.24)	(0.25)
Wisconsin	0.86	0.87	0.89
	(0.15)	(0.15)	(0.16)
Wyoming	0.89	0.90	0.92
	(0.21)	(0.21)	(0.22)
2003	1.10***	1.09***	1.11***
	(0.03)	(0.03)	(0.04)
2004	1.13***	1.10***	1.00
	(0.04)	(0.04)	(0.04)
2005	1.25***	1.20***	1.02
	(0.06)	(0.05)	(0.04)
2006	1.45***	1.37***	1.14***
	(0.07)	(0.07)	(0.06)

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table A4. Cox Proportional Hazard, Exponential, Weibull Survival Model Results (Coefficients and Accelerated Time Form)

Owner Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Female	0.06***	-0.06***	-0.04***
	(0.02)	(0.02)	(0.01)
Mexican	0.02	-0.02	-0.01
	(0.03)	(0.03)	(0.02)
Cuban	-0.00	0.01	0.01
	(0.04)	(0.04)	(0.02)
Puerto Rican	0.18***	-0.18***	-0.12***
	(0.05)	(0.05)	(0.03)
High school	-0.01	0.00	-0.01
	(0.04)	(0.04)	(0.02)
Some college	-0.00	-0.00	-0.01
	(0.04)	(0.04)	(0.03)
Technical or trade school	-0.04	0.04	0.02
	(0.05)	(0.05)	(0.03)
Associates degree	-0.02	0.01	-0.00
	(0.06)	(0.06)	(0.04)
Bachelor's degree	0.01	-0.01	-0.02
	(0.04)	(0.04)	(0.03)
Graduate degree	-0.04	0.03	0.00
	(0.05)	(0.05)	(0.03)
Business Age	0.02	0.01***	0.06***
	(0.02)	(0.00)	(0.00)
Owner age: 25 to 34	-0.12*	0.10	0.03
	(0.06)	(0.06)	(0.04)
35 to 44	-0.21***	0.21***	0.15***
	(0.06)	(0.06)	(0.04)
45 to 54	-0.20***	0.21***	0.16***
	(0.06)	(0.06)	(0.04)
55 to 64	-0.12**	0.12**	0.08**
	(0.06)	(0.06)	(0.04)
Older than 64	-0.05	0.04	-0.00
	(0.06)	(0.06)	(0.04)

Table A4. (cont'd)

Owner Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Hours: 20 to 39	-0.06	0.05	0.02
	(0.05)	(0.05)	(0.03)
40	-0.05	0.04	0.01
	(0.04)	(0.04)	(0.03)
41 to 59	-0.16***	0.15***	0.08***
	(0.04)	(0.04)	(0.03)
More than 59	-0.11**	0.11**	0.05*
	(0.04)	(0.04)	(0.03)
Primary income	-0.17***	0.18***	0.13***
	(0.03)	(0.03)	(0.02)
Provide goods or services	0.04	-0.04	-0.03*
	(0.03)	(0.03)	(0.02)
Management	0.03	-0.02	-0.01
	(0.03)	(0.03)	(0.02)
Finances	-0.06**	0.06**	0.04**
	(0.02)	(0.02)	(0.02)
Multiple establishments	-0.29***	0.29***	0.18***
	(0.04)	(0.04)	(0.02)
Business type: Husband and wife	0.36***	-0.36***	-0.23***
	(0.07)	(0.07)	(0.04)
Estate	XX	XX	XX
	XX	XX	XX
Coop	-0.09	0.08	0.05
	(0.19)	(0.20)	(0.13)
Nonprofit	0.48**	-0.49**	-0.32**
	(0.20)	(0.20)	(0.14)
Public	XX	XX	XX
	XX	XX	XX
Establishment employment	0.00***	-0.00***	-0.00***
	(0.00)	(0.00)	(0.00)
Establishment payroll	-0.00***	0.00***	0.00***
	(0.00)	(0.00)	(0.00)
Firm employment	-0.00**	0.00**	0.00**
	(0.00)	(0.00)	(0.00)
Firm payroll	0.00***	-0.00***	-0.00***
	(0.00)	(0.00)	(0.00)

Table A4. (cont'd)

Owner Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
State-up capital source: savings	-0.06**	0.06**	0.04**
	(0.03)	(0.03)	(0.02)
Assets	0.00	-0.00	-0.00
	(0.03)	(0.03)	(0.02)
Credit	-0.06	0.05	0.03
	(0.04)	(0.04)	(0.02)
Government loan	-0.02	0.02	0.01
	(0.07)	(0.07)	(0.04)
Bank loan	-0.01	0.01	0.00
	(0.03)	(0.03)	(0.02)
Venture capital	0.04	-0.04	-0.03
	(0.06)	(0.06)	(0.04)
None needed	0.06	-0.07	-0.06*
	(0.05)	(0.05)	(0.03)
Expansion capital source: savings	0.09***	-0.10***	-0.08***
	(0.03)	(0.03)	(0.02)
Assets	0.06	-0.06	-0.05*
	(0.04)	(0.04)	(0.03)
Credit	0.05	-0.05	-0.03
	(0.04)	(0.04)	(0.02)
Government loan	0.00	0.01	0.01
	(0.08)	(0.08)	(0.05)
Bank loan	-0.20***	0.22***	0.16***
	(0.04)	(0.04)	(0.02)
Venture capital	0.09	-0.09	-0.07
	(0.07)	(0.07)	(0.05)
None needed	-0.04	0.04	0.03
	(0.04)	(0.04)	(0.02)
>10% sales: federal government	0.10*	-0.10*	-0.06*
	(0.05)	(0.05)	(0.03)
State or local government	-0.17***	0.18***	0.11***
	(0.04)	(0.04)	(0.03)
Primary customer: other businesses	-0.07***	0.07***	0.04***
	(0.03)	(0.03)	(0.02)
Individuals	-0.16***	0.16***	0.10***
	(0.02)	(0.02)	(0.02)

Table A4. (cont'd)

County Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Hispanic establishments	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
Mexican establishments	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
Puerto Rican establishments	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)
Cuban establishments	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)
Hispanic employment	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
Mexican employment	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)
Puerto Rican employment	0.00	-0.00	0.00
	(0.00)	(0.00)	(0.00)
Cuban employment	0.00***	-0.00***	-0.00***
	(0.00)	(0.00)	(0.00)
Interaction: Mexican & Mexican population	-0.00	0.00	-0.00
	(0.00)	(0.00)	(0.00)
Puerto Rican & Puerto Rican population	-0.00**	0.00**	0.00**
	(0.00)	(0.00)	(0.00)
Cuban & Cuban population	-0.00**	0.00**	0.00***
	(0.00)	(0.00)	(0.00)
Mexican & Mexican establishments	-0.00*	0.00*	0.00**
	(0.00)	(0.00)	(0.00)
Puerto Rican & Puerto Rican establishments	0.00	-0.00	0.00
	(0.00)	(0.00)	(0.00)
Cuban & Cuban establishments	0.00	-0.00	0.00
	(0.00)	(0.00)	(0.00)
Mexican & Mexican employment	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)
Puerto Rican & Puerto Rican employment	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
Cuban & Cuban employment	0.00	-0.00	-0.00*
	(0.00)	(0.00)	(0.00)
Population	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)
Population squared	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
Population density	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)

Table A4. (cont'd)

County Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
25 to 34 population	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)
35 to 44 population	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
45 to 54 population	-0.00	0.00	-0.00
	(0.00)	(0.00)	(0.00)
55 to 64 population	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
> 64 population	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
Female population	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
Female population squared	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)
Black population	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)
Black population squared	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
Asian population	0.00*	-0.00*	-0.00*
	(0.00)	(0.00)	(0.00)
Asian population squared	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
Hispanic population	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
Hispanic population squared	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)
Mexican population	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)
Mexican population squared	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)
Puerto Rican population	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
Puerto Rican population squared	0.00***	-0.00***	-0.00**
	(0.00)	(0.00)	(0.00)
Cuban population	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
Cuban population squared	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)

Table A4. (cont'd)

County Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
% with: high school	0.94	-0.90	-0.47
	(0.62)	(0.62)	(0.39)
Some college	1.73**	-1.73**	-1.13**
	(0.69)	(0.69)	(0.44)
Associates degree	0.67	-0.59	-0.19
	(1.17)	(1.18)	(0.75)
Bachelor's degree	0.40	-0.37	-0.13
	(0.67)	(0.67)	(0.43)
Graduate degree	0.88	-0.84	-0.47
	(0.78)	(0.78)	(0.49)
Per capita income	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
Mean household size	0.01	-0.01	-0.01
	(0.09)	(0.09)	(0.06)
% of hh headed by single female	0.24	-0.24	-0.12
	(0.61)	(0.62)	(0.39)
Unemployment rate	0.82	-0.74	-0.31
	(1.03)	(1.03)	(0.66)
Labor force participation rate	0.00	0.01	0.03
	(0.33)	(0.33)	(0.21)
Build permits	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)
Industry employment	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)
State industry employment	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)

Table A4. (cont'd)

County Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Amenities	XX	XX	XX
	XX	XX	XX
Rural-urban code 2	-0.06	-0.04*	-0.01
Metro area, 0.25 to 1 million pop.	(0.04)	(0.02)	(0.03)
Rural-urban code 3	0.02	0.02	0.01
Metro area, < 250,000	(0.06)	(0.03)	(0.04)
Rural-urban code 4	-0.01	0.01	-0.05
Adjacent to metro, ≥ 20,000	(0.07)	(0.04)	(0.05)
Rural-urban code 5	-0.00	0.02	-0.01
Not adjacent to metro, ≥ 20,000	(0.10)	(0.05)	(0.07)
Rural-urban code 6	0.06	0.06	0.01
Adjacent to metro, 2,500-19,999	(0.08)	(0.04)	(0.06)
Rural-urban code 7	0.02	0.00	0.00
Not adjacent to metro, 2,500-19,999	(0.09)	(0.05)	(0.06)
Rural-urban code 8	-0.01	-0.02	0.08
Adjacent to a metro, < 2,500	(0.21)	(0.11)	(0.14)
Rural-urban code 9	0.21	0.14	0.05
Not adjacent to a metro, < 2,500	(0.20)	(0.11)	(0.15)

Table. A4 (cont'd)

NAICS Sector Industry	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
21: Mining, Quarrying, and Oil and Gas Extraction	-0.03	-0.00	0.11
	(0.22)	(0.22)	(0.22)
22: Utilities	XX	XX	XX
	XX	XX	XX
23: Construction	-0.08	-0.05	0.04
	(0.11)	(0.11)	(0.11)
31-33: Manufacturing	-0.08	-0.07	-0.01
	(0.11)	(0.11)	(0.11)
42: Wholesale Trade	-0.20*	-0.18	-0.10
	(0.11)	(0.11)	(0.11)
44-45: Retail Trade	-0.06	-0.03	0.04
	(0.11)	(0.11)	(0.11)
48-49: Transportation and Warehousing	-0.05	-0.03	0.04
	(0.11)	(0.11)	(0.11)
51: Information	XX	XX	XX
	XX	XX	XX
52: Finance and Insurance	XX	XX	XX
	XX	XX	XX
53: Real Estate and Rental and Leasing	-0.22*	-0.20*	-0.12
	(0.12)	(0.12)	(0.12)
54: Professional, Scientific, and Technical Services	-0.17	-0.15	-0.11
	(0.11)	(0.11)	(0.11)
55: Management of Companies and Enterprises	0.09	0.10	0.15
	(0.18)	(0.18)	(0.18)
56: Administrative Support and Waste Management and Remediation Services	-0.07	-0.05	0.01
	(0.11)	(0.11)	(0.12)
61: Educational Services	-0.33**	-0.32**	-0.29**
	(0.14)	(0.14)	(0.14)
62: Health Care and Social Assistance	-0.36***	-0.34***	-0.29**
	(0.12)	(0.12)	(0.12)
71: Arts, Entertainment, and Recreation	-0.08	-0.06	-0.00
	(0.13)	(0.13)	(0.13)
72: Accommodation and Food Services	-0.20*	-0.18	-0.12
	(0.11)	(0.11)	(0.11)
81: Other Services	-0.16	-0.14	-0.08
	(0.11)	(0.11)	(0.11)
92: Public Administration	XX	XX	XX
	XX	XX	XX

Table A4. (cont'd)

State Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Arizona	-0.13	0.12	0.06
	(0.15)	(0.15)	(0.10)
Arkansas	-0.17	0.15	0.07
	(0.22)	(0.22)	(0.14)
California	-0.10	0.09	0.05
	(0.17)	(0.17)	(0.11)
Colorado	0.03	-0.05	-0.05
	(0.14)	(0.14)	(0.09)
Connecticut	0.11	-0.13	-0.09
	(0.18)	(0.18)	(0.11)
Delaware	XX	XX	XX
	XX	XX	XX
District of Columbia	0.08	-0.10	-0.11
	(0.18)	(0.18)	(0.12)
Florida	0.03	-0.04	-0.05
	(0.14)	(0.14)	(0.09)
Georgia	-0.06	0.05	0.02
	(0.14)	(0.14)	(0.09)
Idaho	-0.28	0.28	0.18
	(0.20)	(0.20)	(0.13)
Illinois	-0.18	0.16	0.07
	(0.16)	(0.16)	(0.10)
Indiana	-0.14	0.13	0.06
	(0.17)	(0.17)	(0.11)
Iowa	-0.05	0.04	-0.00
	(0.19)	(0.20)	(0.13)
Kansas	-0.12	0.10	0.05
	(0.17)	(0.18)	(0.11)
Kentucky	-0.01	0.01	0.01
	(0.19)	(0.19)	(0.12)
Louisiana	0.04	-0.04	-0.02
	(0.15)	(0.15)	(0.10)
Maine	XX	XX	XX
	XX	XX	XX
Maryland	-0.30**	0.28*	0.16
	(0.15)	(0.15)	(0.10)
Massachusetts	0.07	-0.09	-0.06
	(0.16)	(0.16)	(0.10)
Michigan	0.02	-0.03	-0.03
	(0.15)	(0.15)	(0.10)

Table A4. (cont'd)

State Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Minnesota	0.11	-0.13	-0.11
	(0.17)	(0.17)	(0.11)
Mississippi	0.24	-0.25	-0.17
	(0.20)	(0.20)	(0.13)
Missouri	-0.12	0.10	0.04
	(0.17)	(0.17)	(0.11)
Montana	XX	XX	XX
	XX	XX	XX
Nebraska	0.06	-0.07	-0.05
	(0.19)	(0.19)	(0.12)
Nevada	-0.10	0.09	0.04
	(0.18)	(0.18)	(0.11)
New Hampshire	-0.35	0.33	0.18
	(0.28)	(0.28)	(0.18)
New Jersey	XX	XX	XX
	XX	XX	XX
New Mexico	-0.21	0.20	0.12
	(0.14)	(0.14)	(0.09)
New York	XX	XX	XX
	XX	XX	XX
North Carolina	-0.11	0.10	0.05
	(0.15)	(0.15)	(0.09)
North Dakota	XX	XX	XX
	XX	XX	XX
Ohio	-0.11	0.10	0.04
	(0.15)	(0.15)	(0.10)
Oklahoma	-0.05	0.03	0.01
	(0.16)	(0.16)	(0.10)
Oregon	-0.26	0.25	0.14
	(0.17)	(0.17)	(0.11)
Pennsylvania	-0.07	0.06	0.02
	(0.16)	(0.16)	(0.10)
Rhode Island	0.06	-0.08	-0.07
	(0.23)	(0.23)	(0.15)
South Carolina	-0.21	0.20	0.10
	(0.18)	(0.18)	(0.12)
South Dakota	0.10	-0.12	-0.11
	(0.26)	(0.26)	(0.16)
Tennessee	-0.15	0.15	0.09
	(0.17)	(0.17)	(0.11)

Table A4. (cont'd)

State and Year Variables	(1) Cox Proportional Hazards Model	(2) Exponential Model	(3) Weibull Model
Texas	-0.00	-0.00	0.00
	(0.14)	(0.14)	(0.09)
Utah	-0.12	0.10	0.05
	(0.20)	(0.20)	(0.13)
Vermont	XX	XX	XX
	XX	XX	XX
Virginia	-0.03	0.02	0.00
	(0.14)	(0.15)	(0.09)
Washington	-0.05	0.04	0.02
	(0.15)	(0.15)	(0.10)
West Virginia	0.08	-0.07	-0.05
	(0.23)	(0.23)	(0.15)
Wisconsin	-0.15	0.14	0.07
	(0.17)	(0.17)	(0.11)
Wyoming	-0.12	0.10	0.05
	(0.24)	(0.24)	(0.15)
2003	0.09***	-0.08***	-0.07***
	(0.03)	(0.03)	(0.02)
2004	0.12***	-0.10***	0.00
	(0.04)	(0.04)	(0.02)
2005	0.22***	-0.18***	-0.01
	(0.04)	(0.04)	(0.03)
2006	0.37***	-0.32***	-0.08***
	(0.05)	(0.05)	(0.03)
Constant		1.53**	1.25***
		(0.68)	(0.43)
Observations	~127,000	~127,000	~127,000

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Table A5. Pooled OLS Regression Results: LOB Employment Growth (Omitted Coefficients)

Business and County Variables	(1)	(2)	(3)	(4)	(5)
	Employment Growth				
	1-year	1-year	1-year	2-year	3-year
Business type: Husband and wife	-0.00	0.00	0.00	-0.02	0.02
	(0.02)	(0.02)	(0.02)	(0.03)	(0.06)
Estate	XX	XX	XX	XX	XX
	XX	XX	XX	XX	XX
Coop	-0.02	-0.04	-0.05	-0.06	-0.02
	(0.06)	(0.06)	(0.06)	(0.08)	(0.12)
Nonprofit	-0.08	-0.08	-0.08	-0.07	-0.10
	(0.08)	(0.08)	(0.08)	(0.11)	(0.17)
Public	XX	XX	XX	XX	XX
	XX	XX	XX	XX	XX
Establishment employment	0.00***	0.00***	0.00***	0.00***	0.00***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Establishment payroll	-0.00	-0.00	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Firm employment	0.00***	0.00***	0.00***	0.00***	0.00***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Firm payroll	-0.00***	-0.00***	-0.00***	-0.00***	-0.00*
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
County-level variables:					
Hispanic establishments	0.00	0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Mexican establishments	-0.00	-0.00	-0.00	-0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Puerto Rican establishments	0.00	-0.00	-0.00	0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Cuban establishments	-0.00	-0.00	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Hispanic employment	0.00	0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Mexican employment	0.00	0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Puerto Rican employment	-0.00	-0.00	-0.00	-0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Cuban employment	-0.00**	-0.00***	-0.00***	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)

Table A5. (cont'd)

County Variables	(1)	(2)	(3)	(4)	(5)
	Employment Growth				
	1-year	1-year	1-year	2-year	3-year
Interaction: Mexican & Mexican population	0.00	0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Puerto Rican & Puerto Rican population	0.00	0.00	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Cuban & Cuban population	0.00	0.00	0.00	-0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Mexican & Mexican establishments	-0.00	-0.00	-0.00	0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Puerto Rican & Puerto Rican establishments	-0.00	-0.00	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Cuban & Cuban establishments	-0.00	-0.00	-0.00	0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Mexican & Mexican employment	0.00	-0.00	-0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Puerto Rican & Puerto Rican employment	0.00	0.00	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Cuban & Cuban employment	-0.00	-0.00	-0.00	-0.00	-0.00*
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Population	0.00	0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Population squared	0.00	0.00	0.00	0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Population density	-0.00	-0.00	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
25 to 34 population	-0.00	-0.00	-0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
35 to 44 population	-0.00	0.00	0.00	-0.00*	-0.00**
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
45 to 54 population	0.00	0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
55 to 64 population	0.00	0.00	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
> 64 population	0.00	0.00	0.00	-0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)

Table A5. (cont'd)

County Variables	(1)	(2)	(3)	(4)	(5)
	Employment Growth				
	1-year	1-year	1-year	2-year	3-year
Female population	-0.00	-0.00	-0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Female population squared	-0.00	-0.00	-0.00	-0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Black population	-0.00	-0.00	-0.00	-0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Black population squared	0.00	0.00	0.00	0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Asian population	-0.00	-0.00	-0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Asian population squared	0.00	0.00	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Hispanic population	-0.00	-0.00	-0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Hispanic population squared	0.00	0.00	0.00	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Mexican population	0.00	0.00	0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Mexican population squared	-0.00	-0.00	-0.00	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Puerto Rican population	0.00**	0.00***	0.00***	0.00	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Puerto Rican population squared	-0.00**	-0.00***	-0.00**	-0.00	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Cuban population	0.00	0.00	0.00	-0.00*	-0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Cuban population squared	0.00	0.00	0.00	0.00*	0.00
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)

Table A5. (cont'd)

State Variables	(1)	(2)	(3)	(4)	(5)
	Employment Growth				
	1-year	1-year	1-year	2-year	3-year
Arizona		0.03	0.04	-0.01	0.01
		(0.04)	(0.04)	(0.06)	(0.08)
Arkansas		-0.00	0.01	-0.10	-0.12
		(0.05)	(0.05)	(0.11)	(0.17)
California		0.05	0.05	-0.05	-0.15
		(0.05)	(0.05)	(0.07)	(0.11)
Colorado		0.03	0.04	-0.06	-0.07
		(0.03)	(0.04)	(0.05)	(0.08)
Connecticut		-0.05	-0.04	-0.09	-0.10
		(0.05)	(0.05)	(0.07)	(0.11)
Delaware		0.01	0.02	0.03	0.07
		(0.05)	(0.05)	(0.07)	(0.11)
District of Columbia		0.03	0.04	-0.00	-0.08
		(0.04)	(0.04)	(0.07)	(0.11)
Florida		0.03	0.04	-0.03	-0.10
		(0.04)	(0.04)	(0.06)	(0.09)
Georgia		0.04	0.04	0.07	0.10
		(0.04)	(0.04)	(0.07)	(0.10)
Idaho		0.03	0.04	0.05	0.16
		(0.04)	(0.05)	(0.07)	(0.11)
Illinois		0.04	0.05	0.04	0.01
		(0.04)	(0.04)	(0.06)	(0.09)
Indiana		0.01	0.02	0.12	0.20
		(0.04)	(0.04)	(0.08)	(0.13)
Iowa		0.03	0.04	-0.01	0.07
		(0.04)	(0.04)	(0.08)	(0.12)
Kansas		0.02	0.03	-0.02	0.00
		(0.04)	(0.04)	(0.07)	(0.11)
Kentucky		0.03	0.04	0.07	0.14
		(0.05)	(0.05)	(0.08)	(0.13)
Louisiana		0.02	0.03	0.04	0.08
		(0.04)	(0.04)	(0.06)	(0.10)
Maine		XX	XX	XX	XX
		XX	XX	XX	XX
Maryland		0.08**	0.09**	0.09	0.11
		(0.04)	(0.04)	(0.06)	(0.09)
Massachusetts		-0.00	-0.01	-0.05	-0.03
		(0.04)	(0.04)	(0.06)	(0.09)
Michigan		0.02	0.03	-0.04	-0.02
		(0.04)	(0.04)	(0.06)	(0.09)

Table A5. (cont'd)

State Variables	(1)	(2)	(3)	(4)	(5)
	Employment Growth				
	1-year	1-year	1-year	2-year	3-year
Minnesota		0.07	0.08*	-0.01	-0.02
		(0.05)	(0.05)	(0.08)	(0.11)
Mississippi		0.00	0.02	0.02	0.14
		(0.06)	(0.06)	(0.11)	(0.10)
Missouri		0.01	0.01	0.01	0.00
		(0.04)	(0.04)	(0.06)	(0.09)
Montana		0.15**	0.16**	0.24*	0.34
		(0.06)	(0.06)	(0.14)	(0.21)
Nebraska		0.04	0.05	0.05	0.09
		(0.05)	(0.05)	(0.09)	(0.13)
Nevada		0.02	0.03	-0.00	0.00
		(0.05)	(0.05)	(0.07)	(0.11)
New Hampshire		0.07	0.08	0.03	0.05
		(0.07)	(0.07)	(0.10)	(0.15)
New Jersey		0.00	0.01	-0.02	-0.05
		(0.03)	(0.04)	(0.05)	(0.08)
New Mexico		0.03	0.04	0.02	0.05
		(0.03)	(0.04)	(0.05)	(0.08)
New York		0.02	0.03	-0.01	-0.01
		(0.04)	(0.05)	(0.07)	(0.11)
North Carolina		0.06*	0.06*	0.00	0.05
		(0.03)	(0.04)	(0.05)	(0.08)
North Dakota		0.09	0.10	0.05	-0.17
		(0.08)	(0.08)	(0.14)	(0.25)
Ohio		0.04	0.04	0.03	0.05
		(0.04)	(0.04)	(0.06)	(0.10)
Oklahoma		0.05	0.06	0.10*	0.13
		(0.03)	(0.04)	(0.06)	(0.09)
Oregon		0.07*	0.07*	0.03	0.06
		(0.04)	(0.04)	(0.06)	(0.09)
Pennsylvania		-0.02	-0.01	-0.02	-0.00
		(0.04)	(0.04)	(0.06)	(0.09)
Rhode Island		0.04	0.05	-0.07	-0.04
		(0.06)	(0.06)	(0.09)	(0.13)
South Carolina		0.09**	0.10**	0.05	0.08
		(0.04)	(0.04)	(0.06)	(0.09)
South Dakota		0.14*	0.15*	-0.06	-0.14
		(0.08)	(0.09)	(0.14)	(0.25)
Tennessee		0.02	0.03	0.00	0.03
		(0.03)	(0.04)	(0.05)	(0.08)

Table A5. (cont'd)

State and Year Variables	(1)	(2)	(3)	(4)	(5)
	Employment Growth				
	1-year	1-year	1-year	2-year	3-year
Texas		0.04	0.04	0.03	0.03
		(0.03)	(0.04)	(0.05)	(0.08)
Utah		0.07	0.08	-0.03	-0.01
		(0.05)	(0.06)	(0.08)	(0.12)
Vermont		0.08	0.09	0.05	0.20
		(0.08)	(0.08)	(0.14)	(0.20)
Virginia		-0.01	-0.00	-0.01	-0.05
		(0.03)	(0.04)	(0.06)	(0.09)
Washington		0.04	0.05	-0.03	0.02
		(0.04)	(0.04)	(0.05)	(0.08)
West Virginia		-0.09	-0.10	-0.18	-0.21
		(0.11)	(0.11)	(0.15)	(0.20)
Wisconsin		0.01	0.02	0.01	0.02
		(0.04)	(0.04)	(0.06)	(0.10)
Wyoming		0.21**	0.22***	0.10	0.19
		(0.08)	(0.09)	(0.14)	(0.19)
2004	-0.17***	-0.17***	-0.17***		
	(0.01)	(0.01)	(0.01)		
2005	-0.17***	-0.17***	-0.17***	-0.16***	
	(0.01)	(0.02)	(0.02)	(0.01)	
2006	-0.14***	-0.14***	-0.14***	-0.15***	-0.13***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
2007	-0.15***	-0.15***	-0.14***	-0.14***	-0.13***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1

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